Prompt Life v0.28.10 Journey Exercise Review

Generated: 2026-06-15T18:46:21.572Z

Readiness judgment: ready for small human testing

Exercises inventoried: 39

Type distribution: tap-to-cycle: 11, sort / group: 8, comparison toggle: 5, single choice: 4, token labeling: 3, connect node: 2, ordered sequence: 2, card stack / tray: 1, mixed / other: 1, multi-select: 2

Priority before: P0: 0, P1: 11, P2: 5, P3: 23

Priority after: P0: 0, P1: 0, P2: 7, P3: 32

Recommendation counts: keep: 39

Visual overflow status: pass

Fixed in this pass: Changed ambiguous “card” wording to “item” in the Journey copy and live context-tray feedback. Added explicit “Choose all” language for the multi-select governance levers. Added explicit “Choose all” language for the multi-select prompt-part picker. Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

Top 10 Exercise Issues / Watch Items

1. What Is an LLM?: watch with testers for clarity/redundancy.
2. Vectors: watch with testers for clarity/redundancy.
3. Tensors: watch with testers for clarity/redundancy.
4. Sampling: watch with testers for clarity/redundancy.
5. Collective Intelligence, Extracted: watch with testers for clarity/redundancy.
6. Benefits Worth Taking Seriously: watch with testers for clarity/redundancy.
7. Model Literate Synthesis: watch with testers for clarity/redundancy.

1. What Is an LLM?

Stage: Before Morning

Learning objective: A large language model is a learned system that turns current context into probabilities for the next token. Practice boundary: An LLM is a conscious reader or a database.

Exercise: Tap through one prompt `tap-to-cycle` `before P2` `after P2`

Prompt: Context enters, a next-token cloud appears, floor is chosen, then the appended response becomes the next context.

Expected action: Tap through context, next-token cloud, chosen token, and updated context.

Items / choices / nodes / steps: context enters | next-token cloud | chosen token | updated context

Target behavior: Finish the prompt-to-next-context sequence.

Feedback behavior: Step-specific insight text appears after each tap.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The prompt life-cycle sequence matters.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

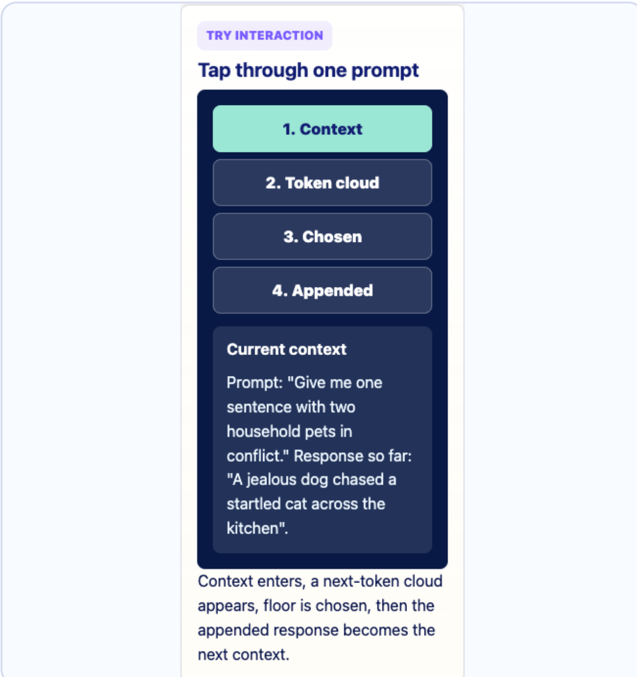
Visual duplicate: Watch during human testing; it may reinforce the visual closely.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.1/5

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.



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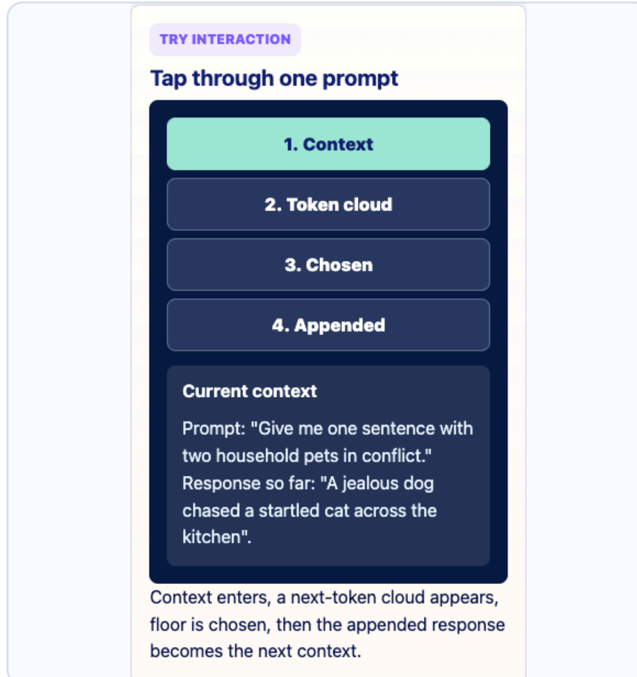

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2. Where LLMs Fit

Stage: Before Morning

Learning objective: A large language model is one kind of generative AI system, built with deep-learning methods inside the broader field of artificial intelligence. Practice boundary: All AI is an LLM, or all generative AI works like ChatGPT.

Exercise: Tap a branch `tap-to-cycle` `before P3` `after P3`

Prompt: Reveal what each branch means in the AI family tree.

Expected action: Tap branches in the AI family tree.

Items / choices / nodes / steps: AI | machine learning | deep learning | generative AI | LLMs | diffusion | multimodal

Target behavior: Reveal at least one branch explanation.

Feedback behavior: Branch-specific one-sentence explanation.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Branches preserve the family-tree structure.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

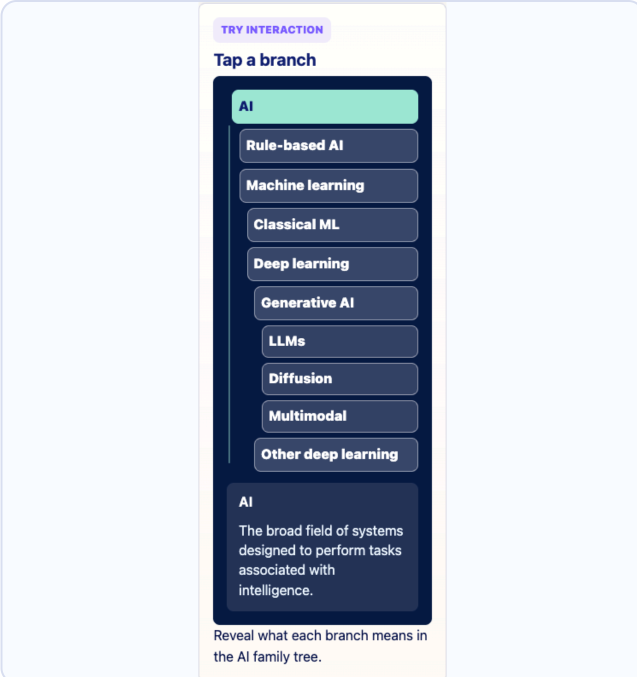
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

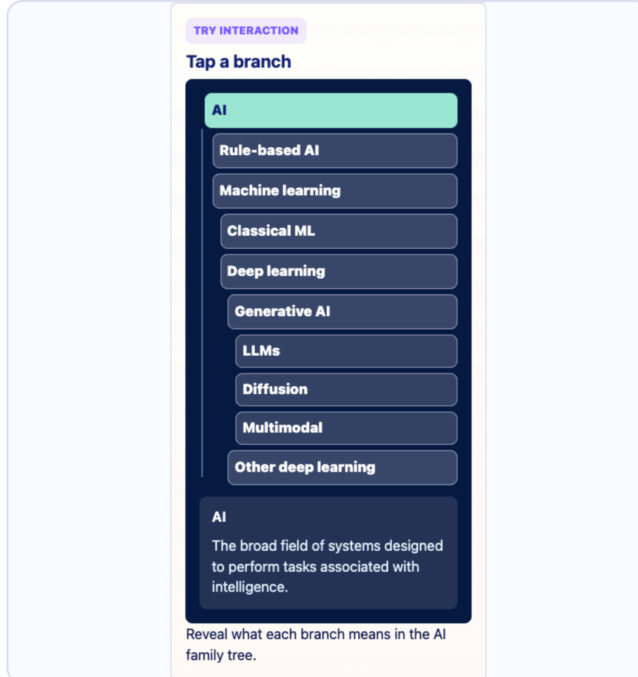
Rubric average: 4.8/5

No P0/P1 issue remains.



TRY INTERACTION
Tap a branch
AI
Rule-based AI
Machine learning
Classical ML
Deep learning
Generative AI
LLMs
Diffusion
Multimodal
Other deep learning
AI
The broad field of systems designed to perform tasks associated with intelligence.
Reveal what each branch means in the AI family tree.

320px



TRY INTERACTION
Tap a branch
AI
Rule-based AI
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The broad field of systems designed to perform tasks associated with intelligence.
Reveal what each branch means in the AI family tree.

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3. Rationalists vs Empiricists

Stage: Before Morning

Learning objective: Symbolic AI tries to reason with explicit rules and symbols; deep learning learns useful patterns from data by adjusting weights. Practice boundary: LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

Exercise: Sort rules and patterns `sort / group` `before P1` `after P3`

Prompt: Some ingredients are explicit rules, some are learned patterns, and many real systems combine both.

Expected action: Tap each item to place it in rules, learned patterns, or hybrid systems.

Items / choices / nodes / steps: explicit rule | learned weights | retrieval wrapper | policy filter

Target behavior: All items land in the best group.

Feedback behavior: Correctness highlighting plus summary insight.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed items keep the rules-versus-patterns contrast compact.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

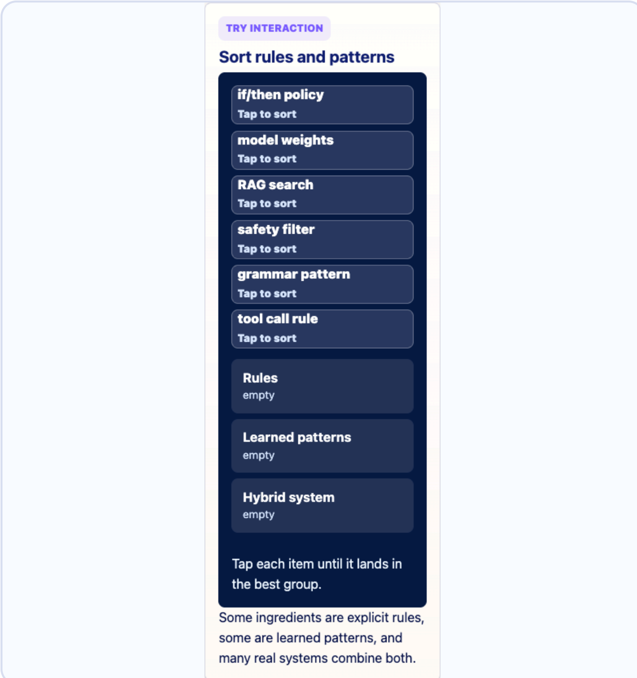
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.



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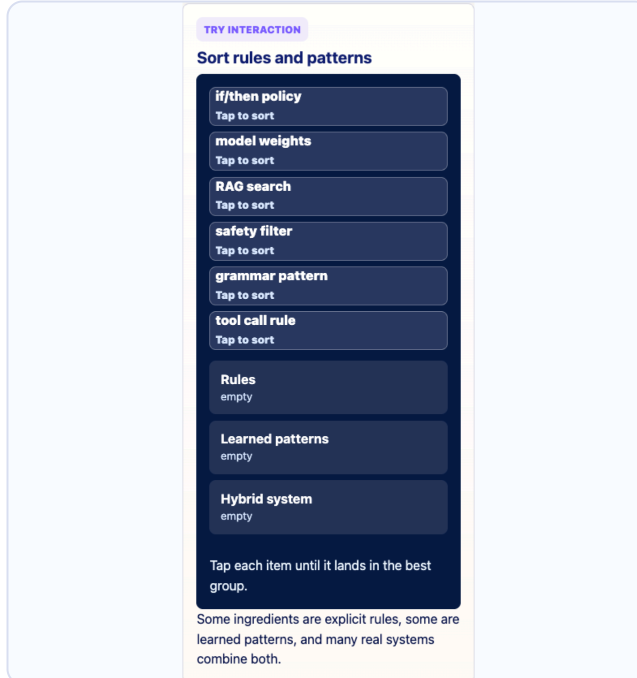

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4. Training

Stage: Before Morning

Learning objective: Training is the process that adjusts model weights by comparing predictions with target answers, such as actual next tokens. Practice boundary: When I chat with the model, it automatically trains itself.

Exercise: Tap through the training loop **tap-to-cycle** **before P3** **after P3**

Prompt: Predict, compare, loss, update weights. The update is the durable learning step.

Expected action: Tap through the training loop.

Items / choices / nodes / steps: predict | compare | loss | update weights

Target behavior: Reach the durable update step.

Feedback behavior: Durable-update explanation.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Training order is the mechanism being taught.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

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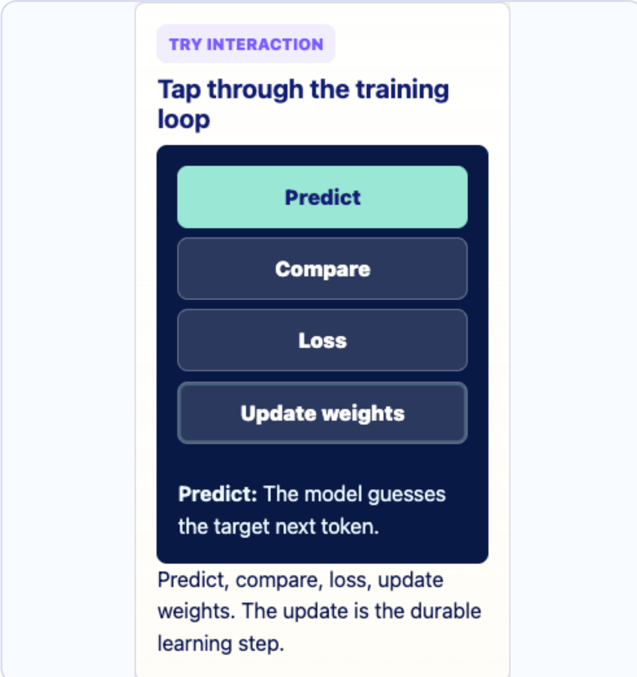
Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Tap through the training loop
Predict
Compare
Loss
Update weights
Predict: The model guesses the target next token.
Predict, compare, loss, update weights. The update is the durable learning step.

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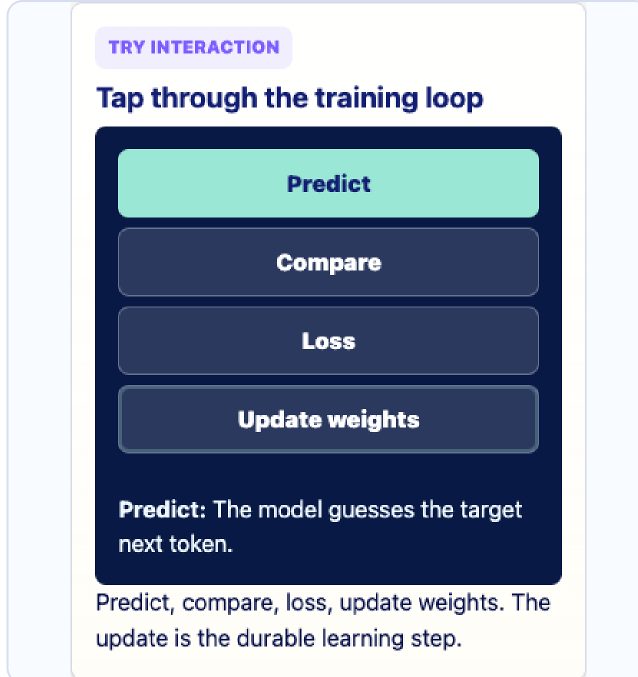
TRY INTERACTION
Tap through the training loop
Predict
Compare
Loss
Update weights
Predict: The model guesses the target next token.
Predict, compare, loss, update weights. The update is the durable learning step.

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5. Pretraining

Stage: Before Morning

Learning objective: Pretraining is the broad first training phase where a model learns general patterns by repeatedly predicting tokens across large datasets. Practice boundary: Pretraining means the model can perfectly recall everything it saw.

Exercise: Pattern learning or recall? comparison toggle before P3 after P3

Prompt: Pretraining is broad durable pattern learning, not a perfect memory of every source.

Expected action: Toggle between broad pattern learning and perfect recall.

Items / choices / nodes / steps: broad pattern learning | perfect recall

Target behavior: Choose broad pattern learning.

Feedback behavior: Boundary feedback against the perfect-memory myth.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Two stable states make the misconception boundary explicit.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

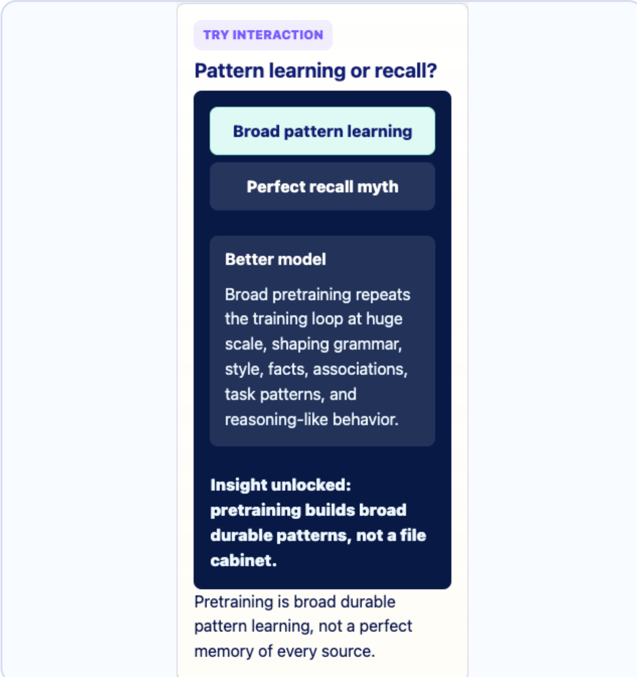
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Pattern learning or recall?
Broad pattern learning
Perfect recall myth
Better model
Broad pretraining repeats the training loop at huge scale, shaping grammar, style, facts, associations, task patterns, and reasoning-like behavior.
Insight unlocked: pretraining builds broad durable patterns, not a file cabinet.
Pretraining is broad durable pattern learning, not a perfect memory of every source.

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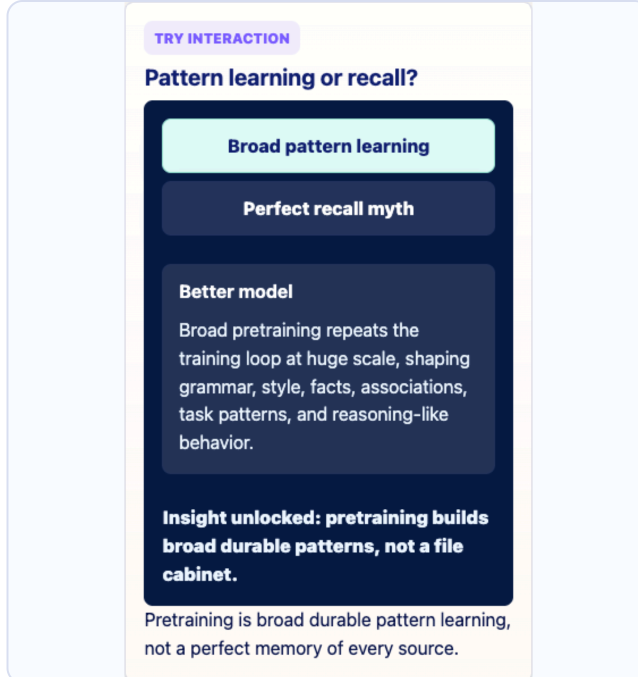
TRY INTERACTION
Pattern learning or recall?
Broad pattern learning
Perfect recall myth
Better model
Broad pretraining repeats the training loop at huge scale, shaping grammar, style, facts, associations, task patterns, and reasoning-like behavior.
Insight unlocked: pretraining builds broad durable patterns, not a file cabinet.
Pretraining is broad durable pattern learning, not a perfect memory of every source.

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6. Overfitting and Generalization

Stage: Before Morning

Learning objective: Overfitting happens when a model learns training examples too narrowly and performs poorly on new examples; generalization means learned patterns work beyond the examples seen during training. Practice boundary: If a model performs well on training data, it has learned well.

Exercise: Choose the better curve `single choice` `before P3` `after P3`

Prompt: The smoother curve may miss a few old dots, but it carries the pattern to validation examples not used in training.

Expected action: Choose the curve that generalizes better.

Items / choices / nodes / steps: training-perfect curve | smoother validation-aware curve

Target behavior: Select the smoother curve.

Feedback behavior: Validation-example feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The side-by-side comparison is spatial.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

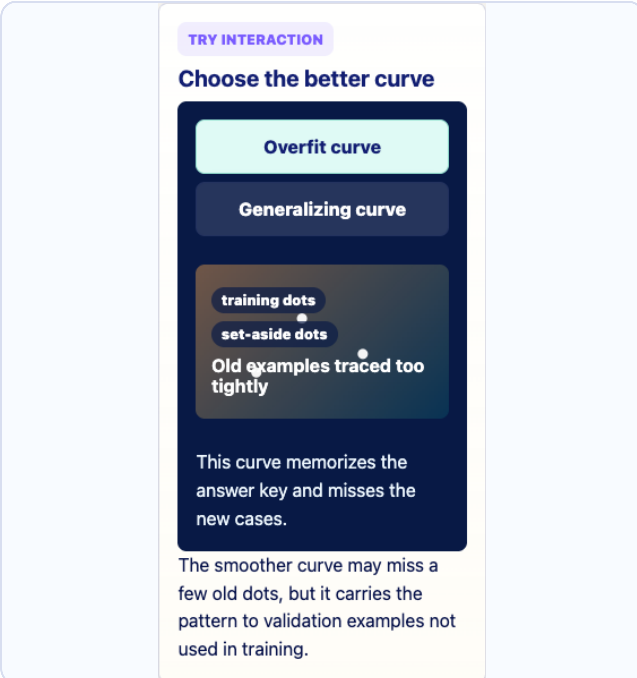
Checkpoint duplicate: Partial risk: watch during human testing so it does not feel like a second checkpoint.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Choose the better curve
Overfit curve
Generalizing curve
training dots
set-aside dots
Old examples traced too tightly
This curve memorizes the answer key and misses the new cases.
The smoother curve may miss a few old dots, but it carries the pattern to validation examples not used in training.

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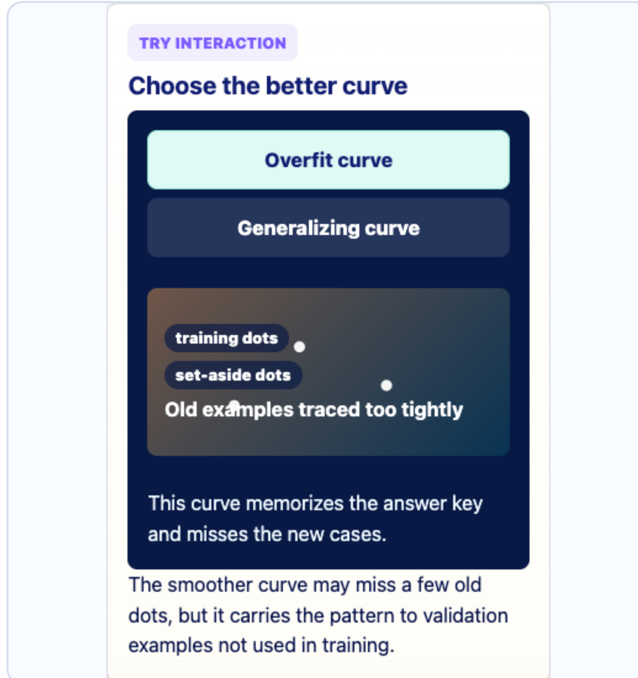
TRY INTERACTION
Choose the better curve
Overfit curve
Generalizing curve
training dots
set-aside dots
Old examples traced too tightly
This curve memorizes the answer key and misses the new cases.
The smoother curve may miss a few old dots, but it carries the pattern to validation examples not used in training.

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7. Fine-Tuning

Stage: Before Morning

Learning objective: Fine-tuning is additional training that adapts a pretrained model toward a task, domain, style, or preference. Practice boundary: Prompting once is the same as fine-tuning.

Exercise: Sort the steering moves `sort / group` `before P1` `after P3`

Prompt: Fine-tuning is durable training. Prompting and RAG steer the current run. Sampling chooses the next token.

Expected action: Sort durable training, temporary context, and decoding moves.

Items / choices / nodes / steps: fine-tuning | prompting | RAG | sampling

Target behavior: Each steering move is assigned to the right change type.

Feedback behavior: Bucket-specific correction.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable ordering keeps nearby terms comparable.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

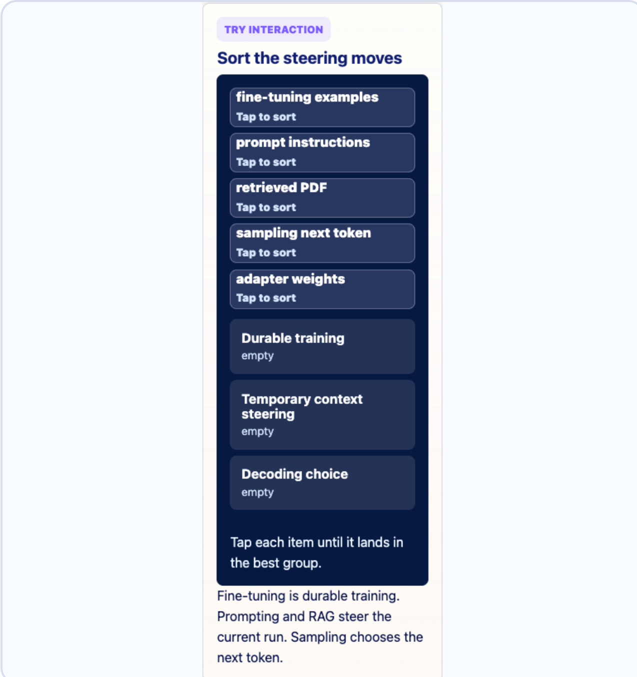
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

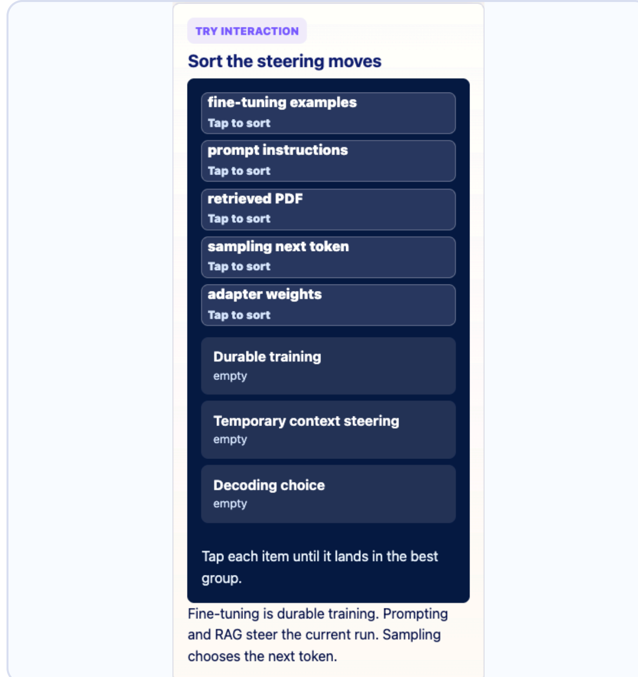
Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.


TRY INTERACTION
Sort the steering moves
fine-tuning examples
Tap to sort
prompt instructions
Tap to sort
retrieved PDF
Tap to sort
sampling next token
Tap to sort
adapter weights
Tap to sort
Durable training
empty
Temporary context steering
empty
Decoding choice
empty
Tap each item until it lands in the best group.
Fine-tuning is durable training. Prompting and RAG steer the current run. Sampling chooses the next token.

320px


TRY INTERACTION
Sort the steering moves
fine-tuning examples
Tap to sort
prompt instructions
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adapter weights
Tap to sort
Durable training
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Temporary context steering
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Decoding choice
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Tap each item until it lands in the best group.
Fine-tuning is durable training. Prompting and RAG steer the current run. Sampling chooses the next token.

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8. Alignment

Stage: Before Morning

Learning objective: Alignment is the effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values. Practice boundary: Aligned means morally good or fully trustworthy.

Exercise: Group the alignment methods **sort / group** **before P1** **after P3**

Prompt: Alignment can use durable shaping, runtime steering, and evaluation. It shapes behavior, not moral agency.

Expected action: Group alignment methods by durable shaping, runtime steering, and evaluation.

Items / choices / nodes / steps: instruction tuning | system prompt | policy filter | safety test

Target behavior: Each method is assigned to its mechanism group.

Feedback behavior: Method-group summary.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable group labels preserve the boundary.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

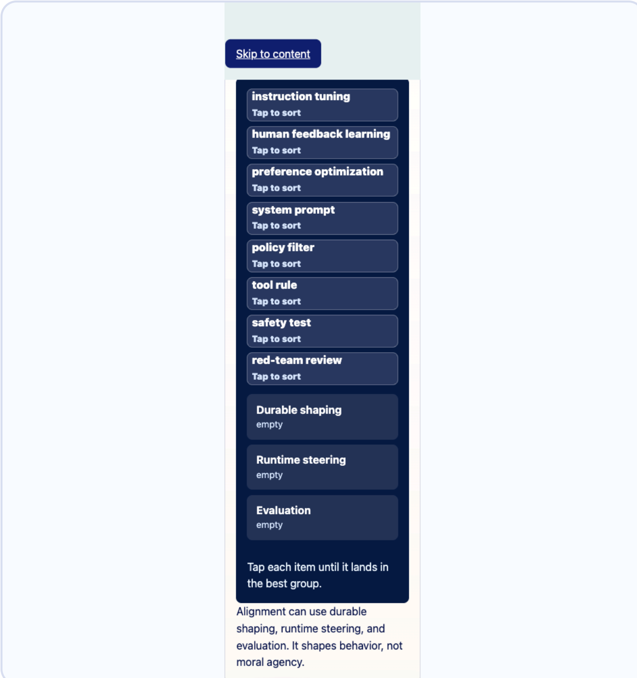
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.



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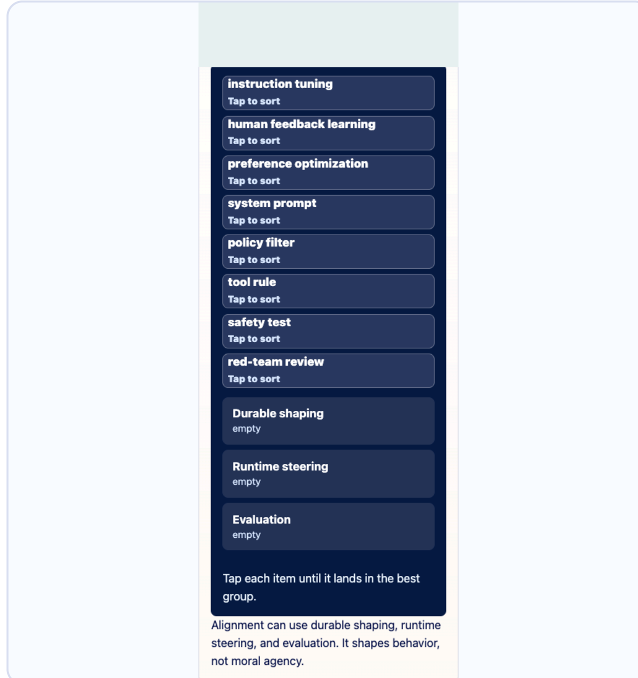

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9. Inference

Stage: Morning Commute

Learning objective: Inference is normal model use: fixed learned weights process the current context to produce scores for the next response token. Practice boundary: Normal chat automatically trains the model.

Exercise: Walk the forward pass tap-to-cycle before P3 after P3

Prompt: The current context moves through the model. Temporary activations change, but durable weights do not.

Expected action: Tap through the forward pass from current context to next token.

Items / choices / nodes / steps: current context | temporary activations | fixed weights | next-token scores

Target behavior: Complete the forward pass without durable weight change.

Feedback behavior: Temporary-versus-durable feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Forward-pass order is the model mechanism.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

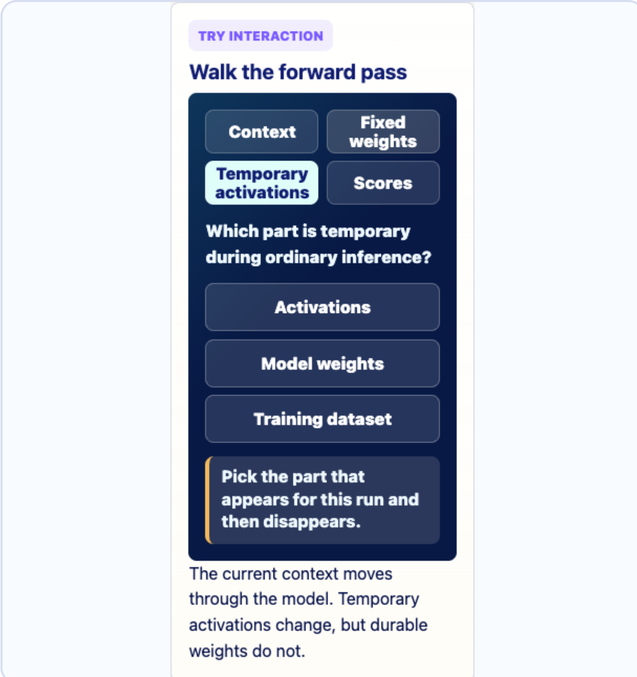
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Walk the forward pass
Context
Fixed weights
Temporary activations
Scores
Which part is temporary during ordinary inference?
Activations
Model weights
Training dataset
Pick the part that appears for this run and then disappears.
The current context moves through the model. Temporary activations change, but durable weights do not.

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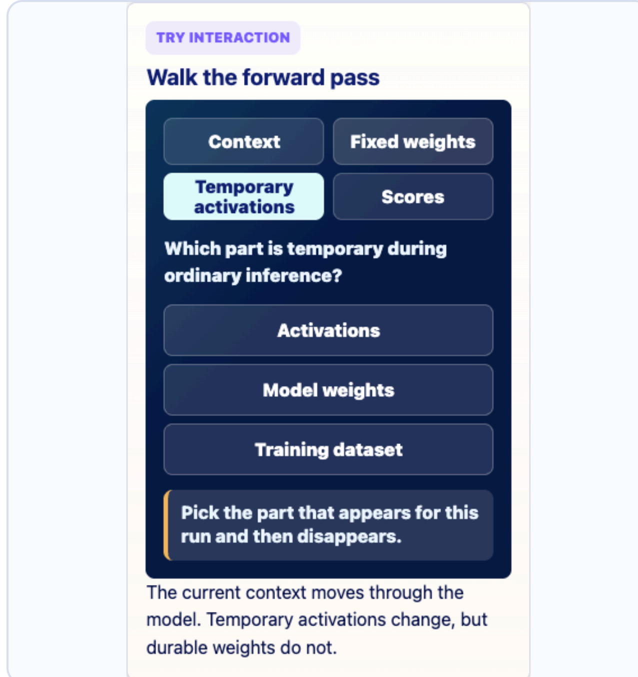
TRY INTERACTION
Walk the forward pass
Context
Fixed weights
Temporary activations
Scores
Which part is temporary during ordinary inference?
Activations
Model weights
Training dataset
Pick the part that appears for this run and then disappears.
The current context moves through the model. Temporary activations change, but durable weights do not.

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10. Prompt vs Response

Stage: Morning Commute

Learning objective: Prompt tokens are given to the model; response tokens are generated by the model one at a time and appended to the context. Practice boundary: The model generates the whole response at once.

Exercise: Label the context pieces **token labeling** **before P3** **after P3**

Prompt: Tap each row to see prompt, response so far, next token, and current context stay distinct.

Expected action: Tap rows that separate prompt, response so far, next token, and updated context.

Items / choices / nodes / steps: given prompt | response so far | next token | updated context

Target behavior: Inspect all four context pieces.

Feedback behavior: Row-specific explanation.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Rows mirror the generation trace.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

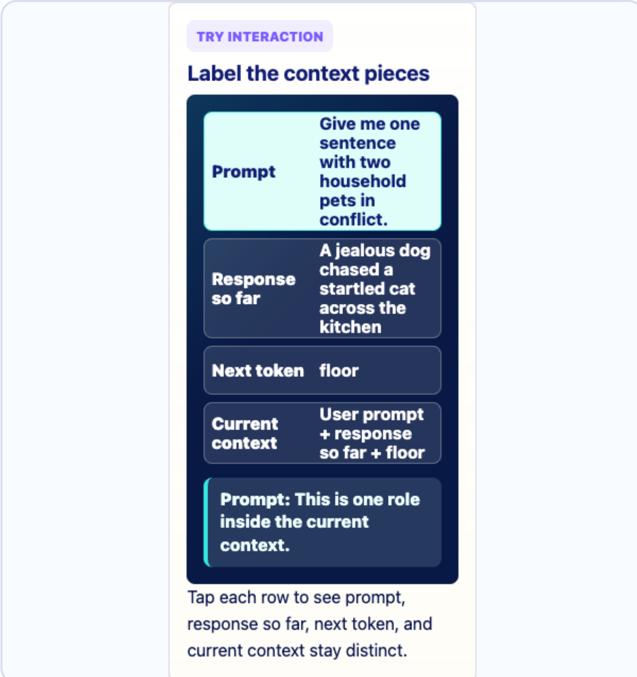
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

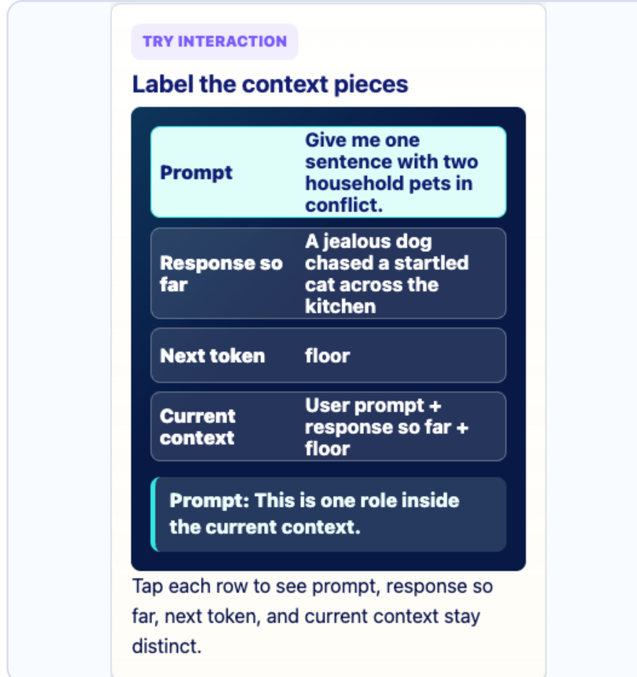
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Label the context pieces
Prompt
Give me one sentence with two household pets in conflict.
Response so far
A jealous dog chased a startled cat across the kitchen
Next token
floor
Current context
User prompt + response so far + floor
Prompt: This is one role inside the current context.
Tap each row to see prompt, response so far, next token, and current context stay distinct.

320px


TRY INTERACTION
Label the context pieces
Prompt
Give me one sentence with two household pets in conflict.
Response so far
A jealous dog chased a startled cat across the kitchen
Next token
floor
Current context
User prompt + response so far + floor
Prompt: This is one role inside the current context.
Tap each row to see prompt, response so far, next token, and current context stay distinct.

390px

11. Tokenization

Stage: Morning Commute

Learning objective: Tokenization splits text into chunks that can be represented as token IDs. Practice boundary: Tokens are always whole words or human concepts.

Exercise: Split the sentence token labeling before P3 after P3

Prompt: Tap once for token pieces, then show uneven chunks and punctuation.

Expected action: Reveal token pieces and uneven chunks.

Items / choices / nodes / steps: A | jealous | dog | chased | start|led | floor|.

Target behavior: Notice tokens can be uneven and include punctuation.

Feedback behavior: Token boundary feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The sentence order makes token boundaries readable.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

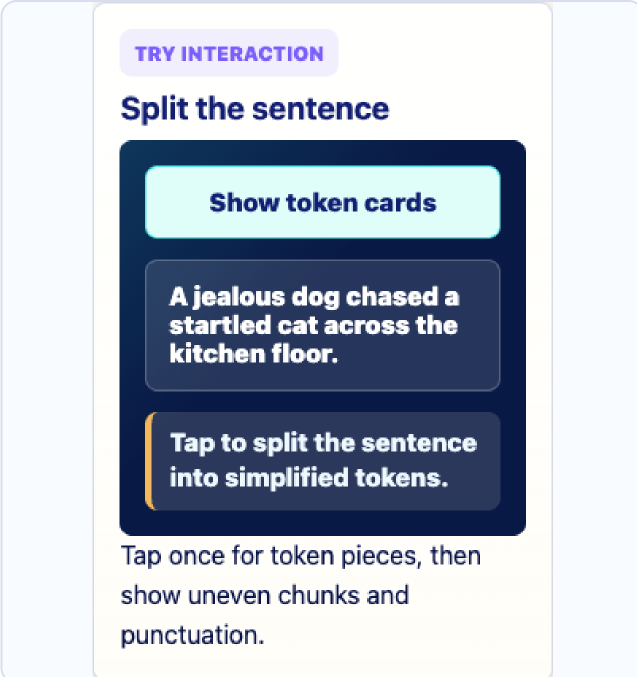
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Split the sentence
Show token cards
A jealous dog chased a startled cat across the kitchen floor.
Tap to split the sentence into simplified tokens.
Tap once for token pieces, then show uneven chunks and punctuation.

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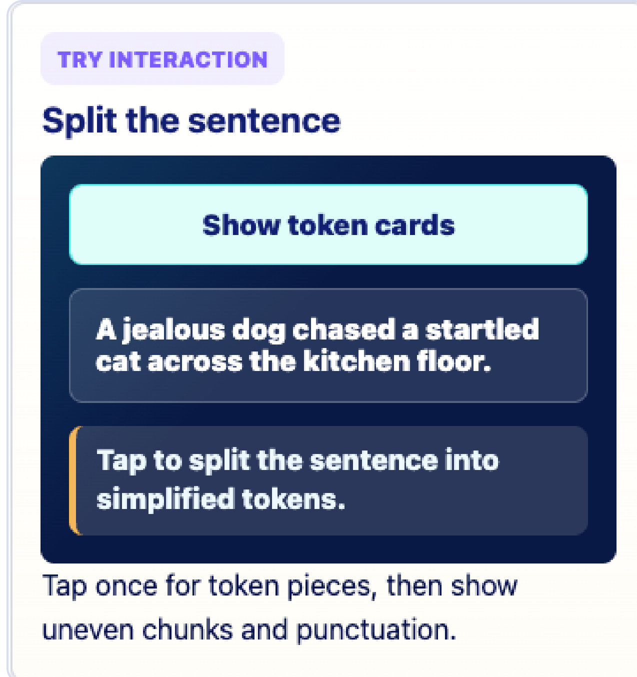
TRY INTERACTION
Split the sentence
Show token cards
A jealous dog chased a startled cat across the kitchen floor.
Tap to split the sentence into simplified tokens.
Tap once for token pieces, then show uneven chunks and punctuation.

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12. Token IDs

Stage: Morning Commute

Learning objective: A token ID is the number assigned to a token so the model can look up its learned starting vector. Practice boundary: The token ID is the meaning.

Exercise: Match token to ID token labeling before P3 after P3

Prompt: Match dog, cat, and floor to their lookup numbers and table rows.

Expected action: Match token text to token IDs and table rows.

Items / choices / nodes / steps: dog | cat | floor | ID rows

Target behavior: Reveal lookup relationship without treating IDs as meaning.

Feedback behavior: ID-is-not-meaning feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable rows teach lookup rather than memory.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

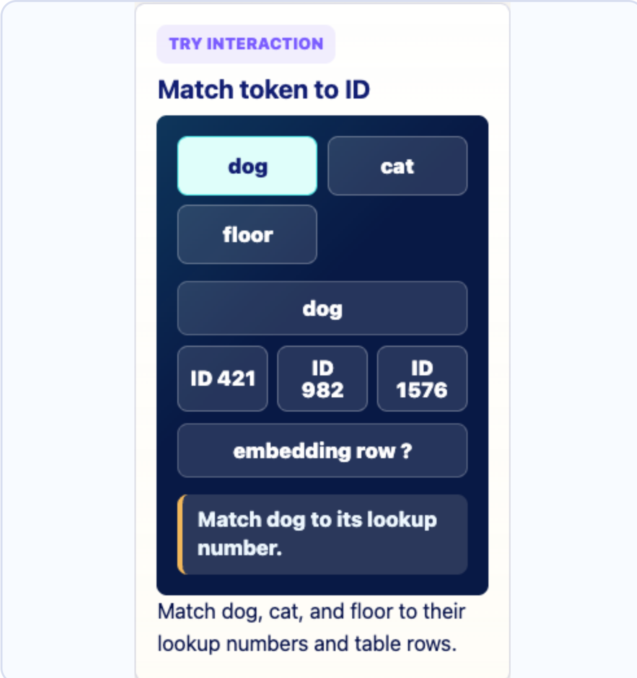
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

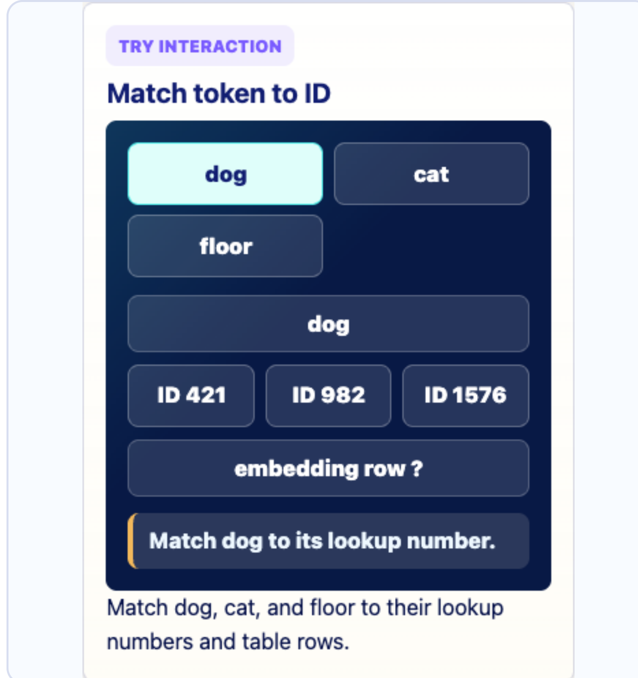
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Match token to ID
dog
cat
floor
dog
ID 421
ID 982
ID 1576
embedding row ?
Match dog to its lookup number.
Match dog, cat, and floor to their lookup numbers and table rows.

320px


TRY INTERACTION
Match token to ID
dog
cat
floor
dog
ID 421
ID 982
ID 1576
embedding row ?
Match dog to its lookup number.
Match dog, cat, and floor to their lookup numbers and table rows.

390px

13. Embeddings

Stage: Morning Commute

Learning objective: An embedding is a learned starting vector retrieved for a token ID. Practice boundary: An embedding is a dictionary definition or a human memory.

Exercise: Look up a starting vector **tap-to-cycle** **before P3** **after P3**

Prompt: Tap a token ID to retrieve its embedding row and notice what is durable versus temporary.

Expected action: Tap a token ID to retrieve its embedding row.

Items / choices / nodes / steps: token ID | embedding table row | starting vector

Target behavior: Show the starting vector before context reshapes it.

Feedback behavior: Durable embedding versus temporary hidden-state feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Lookup order matches the data path.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

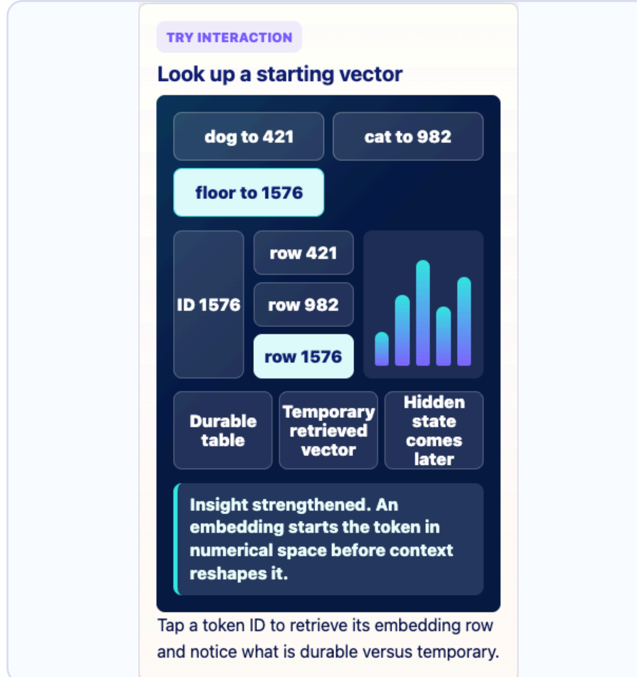
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Look up a starting vector
dog to 421
cat to 982
floor to 1576
ID 1576
row 421
row 982
row 1576
Durable table
Temporary retrieved vector
Hidden state comes later
Insight strengthened. An embedding starts the token in numerical space before context reshapes it.
Tap a token ID to retrieve its embedding row and notice what is

320px


TRY INTERACTION
Look up a starting vector
dog to 421
cat to 982
floor to 1576
ID 1576
row 421
row 982
row 1576
Durable table
Temporary retrieved vector
Hidden state comes later
Insight strengthened. An embedding starts the token in numerical space before context reshapes it.
Tap a token ID to retrieve its embedding row and notice what is durable versus temporary.

390px

14. Vectors

Stage: Morning Commute

Learning objective: A vector is a list of numbers that represents features of something, such as a token. Practice boundary: Each vector dimension has a simple human-readable label.

Exercise: Toggle vector views `comparison toggle` `before P2` `after P2`

Prompt: Compare a simplified teaching view with the messier distributed view real models use.

Expected action: Switch between teaching sliders and distributed feature bars.

Items / choices / nodes / steps: teaching view | distributed view

Target behavior: Reveal why real vector dimensions are distributed.

Feedback behavior: Dimension-label limit feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Two-state comparison keeps the abstraction manageable.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

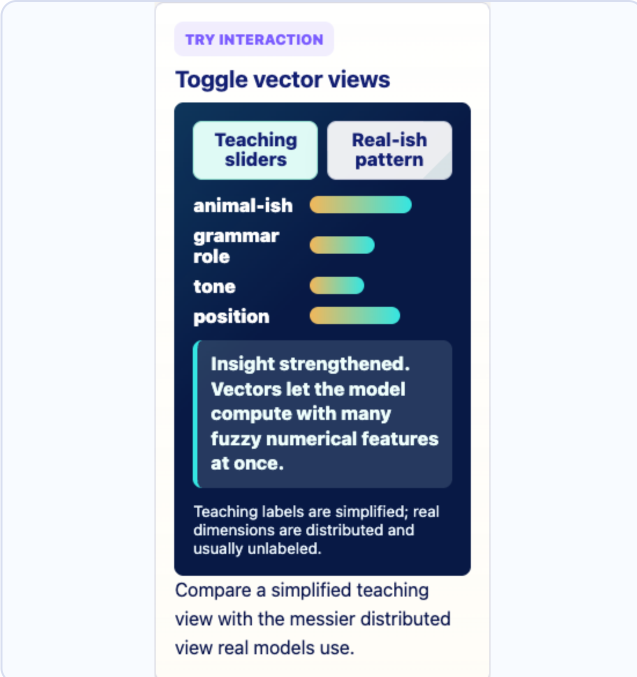
Visual duplicate: Watch during human testing; it may reinforce the visual closely.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

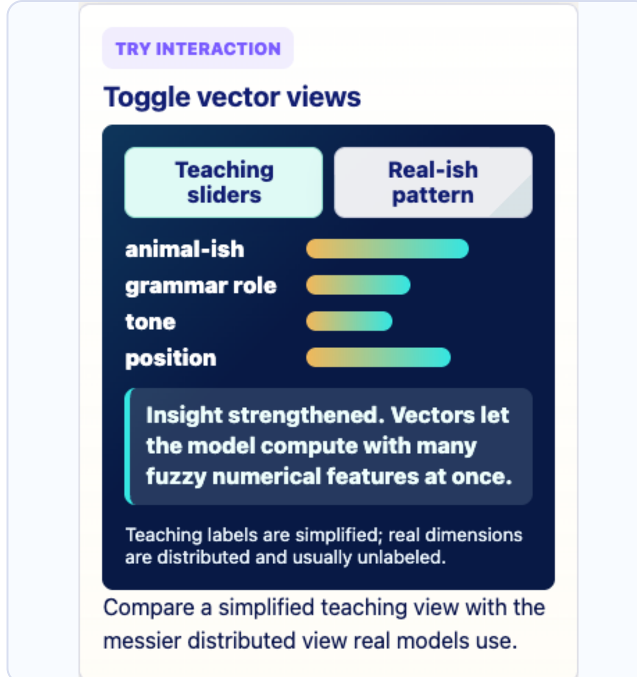
Rubric average: 4.2/5

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.


TRY INTERACTION
Toggle vector views
Teaching sliders
Real-ish pattern
animal-ish
grammar role
tone
position
Insight strengthened. Vectors let the model compute with many fuzzy numerical features at once.
Teaching labels are simplified; real dimensions are distributed and usually unlabeled.
Compare a simplified teaching view with the messier distributed view real models use.

320px


TRY INTERACTION
Toggle vector views
Teaching sliders
Real-ish pattern
animal-ish
grammar role
tone
position
Insight strengthened. Vectors let the model compute with many fuzzy numerical features at once.
Teaching labels are simplified; real dimensions are distributed and usually unlabeled.
Compare a simplified teaching view with the messier distributed view real models use.

390px

15. Tensors

Stage: Morning Commute

Learning objective: A tensor is an organized block of numbers, often holding many token vectors at once. Practice boundary: A tensor is one word, one number, or a mysterious object.

Exercise: Inspect the axes `tap-to-cycle` `before P2` `after P2`

Prompt: Tap token axis, feature axis, and batch note to see how vectors become a shaped block.

Expected action: Inspect token axis, feature axis, and batch note.

Items / choices / nodes / steps: token axis | feature axis | batch note

Target behavior: See vectors as a shaped numerical block.

Feedback behavior: Axis-specific explanation.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Axis order follows the tensor teaching diagram.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

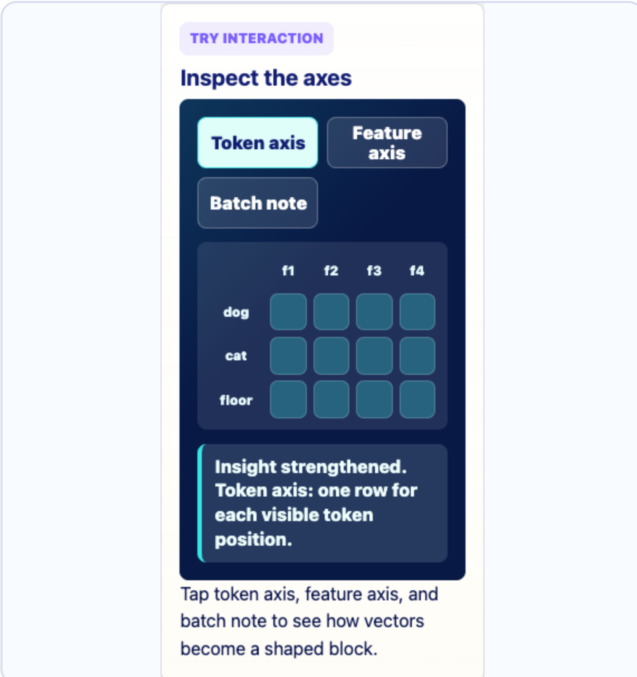
Visual duplicate: Watch during human testing; it may reinforce the visual closely.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

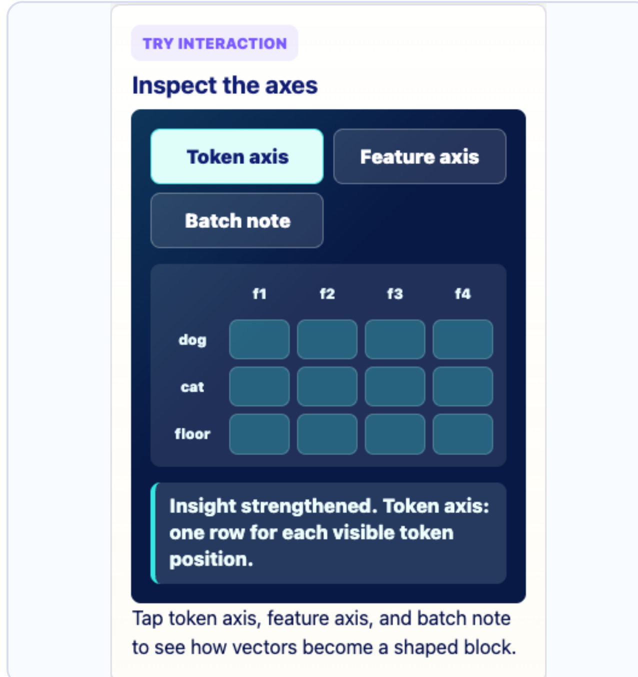
Rubric average: 4.1/5

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value. Uses “batch note” as a legitimate tensor concept, not implementation-process language.



320px



390px

16. Attention

Stage: Workday

Learning objective: Attention is a mechanism that assigns relevance weights between token positions in the current context. Practice boundary: Attention means consciousness or awareness.

Exercise: Connect the pronoun connect node before P3 after P3

Prompt: Choose which earlier token helps interpret “it.” Attention is relevance, not awareness.

Expected action: Connect the pronoun token to the most relevant source token.

Items / choices / nodes / steps: dog | cat | it

Target behavior: Select cat as the likely referent.

Feedback behavior: Relevance-not-awareness feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed sentence order preserves the pronoun example.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

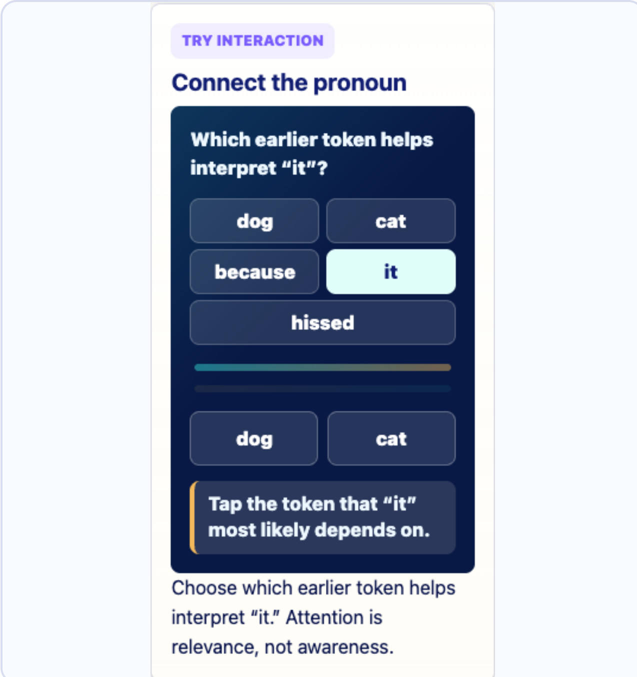
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

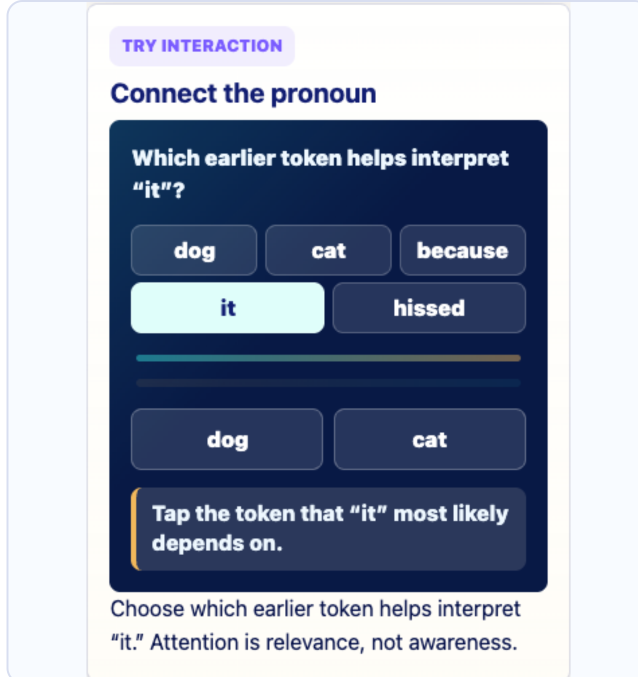
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Connect the pronoun
Which earlier token helps interpret “it”?
dog
cat
because
it
hissed
dog
cat
Tap the token that “it” most likely depends on.
Choose which earlier token helps interpret “it.” Attention is relevance, not awareness.

320px


TRY INTERACTION
Connect the pronoun
Which earlier token helps interpret “it”?
dog
cat
because
it
hissed
dog
cat
Tap the token that “it” most likely depends on.
Choose which earlier token helps interpret “it.” Attention is relevance, not awareness.

390px

17. MLP

Stage: Workday

Learning objective: An MLP, or multi-layer perceptron, is a feed-forward network that reshapes each token’s feature vector inside a transformer layer. Practice boundary: The MLP mixes information between different token positions.

Exercise: Before and after MLP comparison toggle before P3 after P3

Prompt: Toggle the same token position before and after the MLP. The feature vector changes; the token ID and weights do not.

Expected action: Toggle the same token before and after the MLP.

Items / choices / nodes / steps: before MLP | after MLP

Target behavior: Notice feature reshaping without changing the token ID or weights.

Feedback behavior: Feature reshaping feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Before/after comparison is the lesson point.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Before and after MLP
Before MLP
After MLP
cat
same token position
What changed?
The token's feature vector
Permanent memory
The token ID
Toggle before and after, then choose what changed.
Toggle the same token position before and after the MLP. The feature vector changes; the token ID and weights do not.

320px


TRY INTERACTION
Before and after MLP
Before MLP
After MLP
cat
same token position
What changed?
The token's feature vector
Permanent memory
The token ID
Toggle before and after, then choose what changed.
Toggle the same token position before and after the MLP. The feature vector changes; the token ID and weights do not.

390px

18. Layers

Stage: Workday

Learning objective: A layer is a repeated transformer block that refines hidden states through attention, MLPs, residual paths, and normalization. Practice boundary: Each layer is a human-like thought or reasoning step.

Exercise: Inspect the stack **tap-to-cycle** **before P3** **after P3**

Prompt: Tap a layer to watch the same hidden-state ribbon move forward through repeated attention and MLP blocks.

Expected action: Tap layers in a transformer stack.

Items / choices / nodes / steps: layer 1 | layer 2 | layer 3 | layer 4

Target behavior: Inspect repeated attention and MLP blocks.

Feedback behavior: Repeated numerical transformation feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Layer order is sequential.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

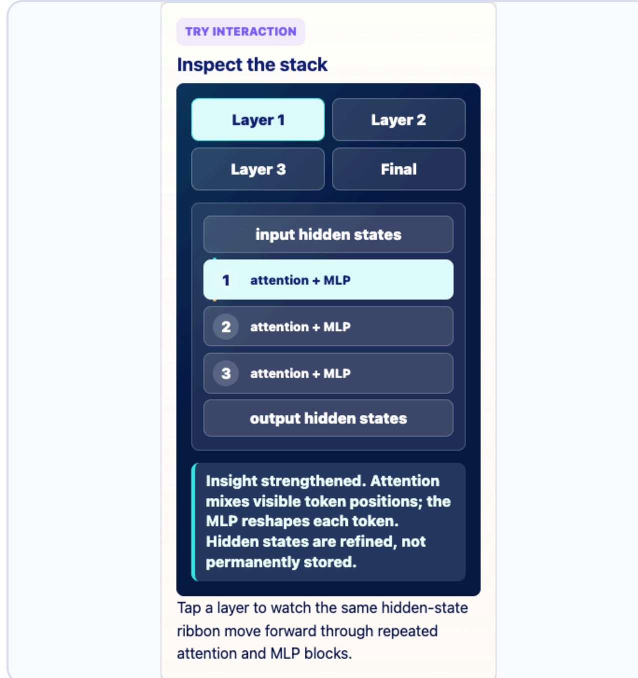
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Inspect the stack
Layer 1
Layer 2
Layer 3
Final
input hidden states
1 attention + MLP
2 attention + MLP
3 attention + MLP
output hidden states
Insight strengthened. Attention mixes visible token positions; the MLP reshapes each token. Hidden states are refined, not permanently stored.
Tap a layer to watch the same hidden-state ribbon move forward through repeated attention and MLP blocks.

320px


TRY INTERACTION
Inspect the stack
Layer 1
Layer 2
Layer 3
Final
input hidden states
1 attention + MLP
2 attention + MLP
3 attention + MLP
output hidden states
Insight strengthened. Attention mixes visible token positions; the MLP reshapes each token. Hidden states are refined, not permanently stored.
Tap a layer to watch the same hidden-state ribbon move forward through repeated attention and MLP blocks.

390px

19. Hidden States

Stage: Workday

Learning objective: A hidden state is a temporary context-shaped vector for a token at a particular layer. Practice boundary: Hidden states are secret memories, encrypted English, or stored thoughts.

Exercise: Durable or temporary? sort / group before P1 after P3

Prompt: Sort model parts by whether they persist as learned parameters, appear temporarily during this run, or sit outside the current forward pass.

Expected action: Sort durable, temporary, and outside-forward-pass items.

Items / choices / nodes / steps: embedding table | weight | hidden state | attention pattern | training data

Target behavior: Separate learned parameters, temporary run state, and outside material.

Feedback behavior: Temporary hidden-state feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed labels preserve the three-way boundary.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

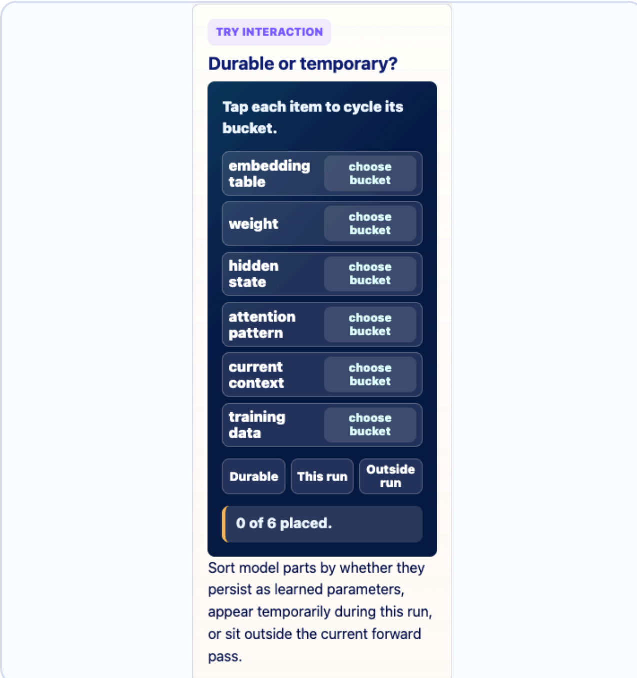
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

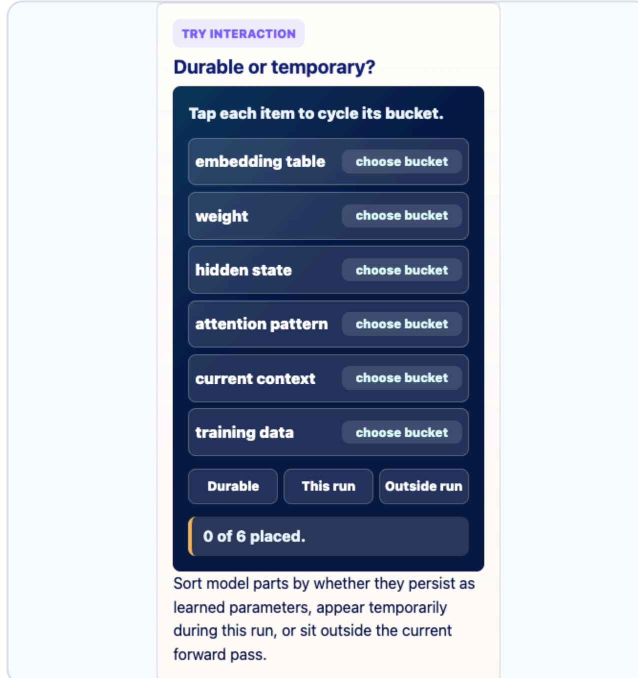
Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.



320px



390px

20. Logits

Stage: Decision Room

Learning objective: Logits are raw scores for possible next tokens before they are turned into probabilities. Practice boundary: Logits are probabilities or truth scores.

Exercise: Raw scores or probabilities? comparison toggle | before P3 | after P3

Prompt: Toggle the display and notice that logits are the before state: raw scores, not probabilities or truth.

Expected action: Toggle raw scores and probability-ready framing.

Items / choices / nodes / steps: raw scores | not probabilities | not truth

Target behavior: Keep logits as raw next-token scores.

Feedback behavior: Logits-are-not-probabilities feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: One contrast is enough at this point in the Journey.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

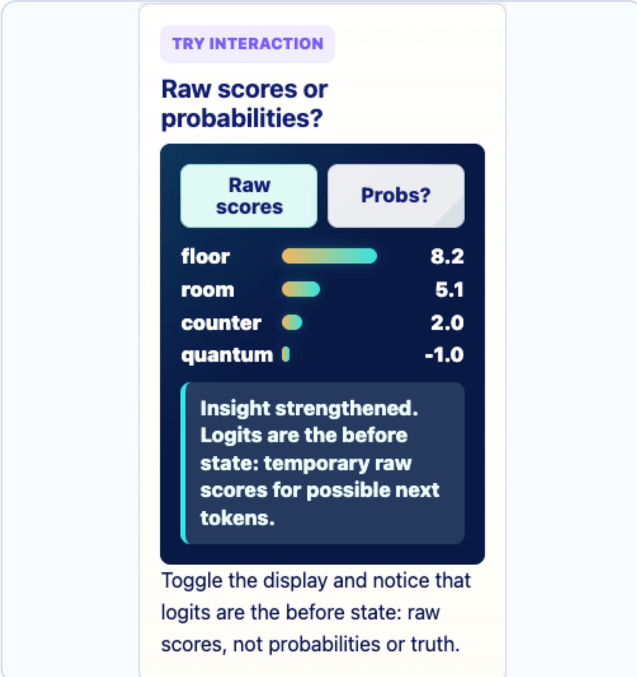
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

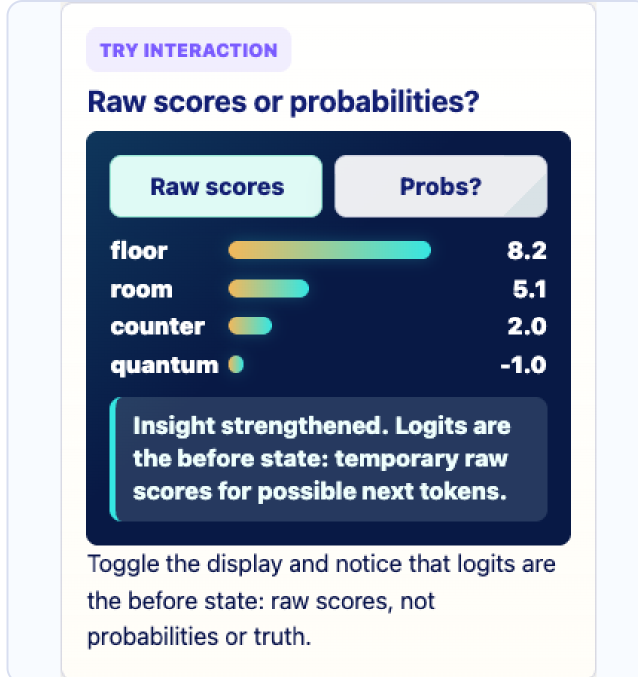
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Raw scores or probabilities?
Raw scores
Probs?
floor 8.2
room 5.1
counter 2.0
quantum -1.0
Insight strengthened. Logits are the before state: temporary raw scores for possible next tokens.
Toggle the display and notice that logits are the before state: raw scores, not probabilities or truth.

320px


TRY INTERACTION
Raw scores or probabilities?
Raw scores
Probs?
floor 8.2
room 5.1
counter 2.0
quantum -1.0
Insight strengthened. Logits are the before state: temporary raw scores for possible next tokens.
Toggle the display and notice that logits are the before state: raw scores, not probabilities or truth.

390px

21. Softmax

Stage: Decision Room

Learning objective: Softmax is a function that turns raw next-token scores into probabilities that sum to one. Practice boundary: Softmax guarantees truth or certainty.

Exercise: Convert the scores tap-to-cycle before P3 after P3

Prompt: Tap softmax to turn raw logits into probabilities that sum to 100%. Probability is still not truth.

Expected action: Convert raw scores into probabilities.

Items / choices / nodes / steps: raw logits | softmax | probabilities that sum to 100%

Target behavior: See probabilities emerge from raw scores.

Feedback behavior: Probability-not-truth feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The transform has a natural before/after order.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

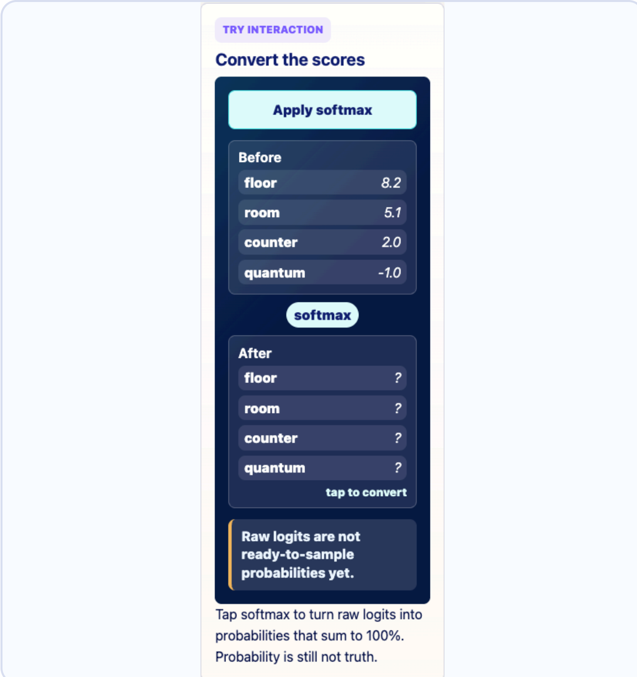
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

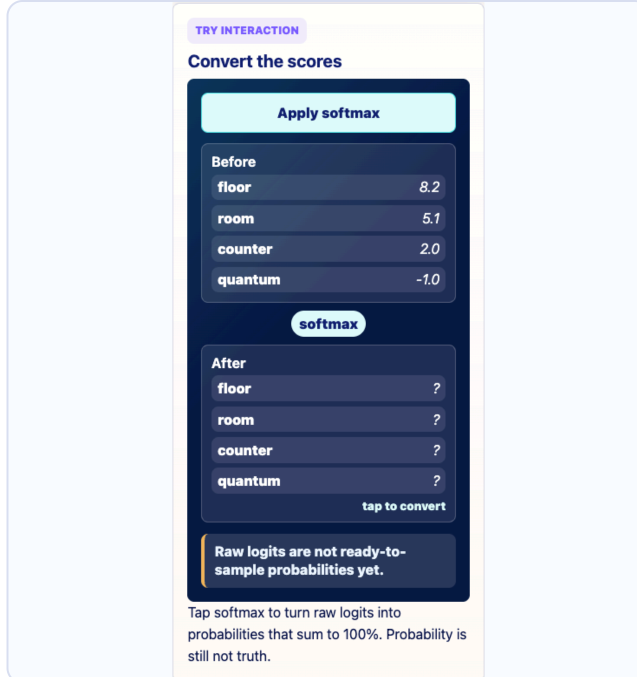
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Convert the scores
Apply softmax
Before
floor 8.2
room 5.1
counter 2.0
quantum -1.0
softmax
After
floor ?
room ?
counter ?
quantum ?
tap to convert
Raw logits are not ready-to-sample probabilities yet.
Tap softmax to turn raw logits into probabilities that sum to 100%. Probability is still not truth.

320px


TRY INTERACTION
Convert the scores
Apply softmax
Before
floor 8.2
room 5.1
counter 2.0
quantum -1.0
softmax
After
floor ?
room ?
counter ?
quantum ?
tap to convert
Raw logits are not ready-to-sample probabilities yet.
Tap softmax to turn raw logits into probabilities that sum to 100%. Probability is still not truth.

390px

22. Sampling

Stage: Decision Room

Learning objective: Sampling is the decoding step that chooses one next token from the probability distribution. Practice boundary: Sampling writes the whole response or permanently teaches the model.

Exercise: Choose from the probability cloud single choice before P2 after P2

Prompt: Pick the most likely next token from the local context. Probability is not truth; a likely token can still be unsupported in other situations.

Expected action: Choose a next token from probability-shaped candidates.

Items / choices / nodes / steps: floor | room | quantum

Target behavior: Pick a likely next token while keeping probability separate from truth.

Feedback behavior: Sampling uncertainty feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed candidates keep the example short.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: Partial risk: watch during human testing so it does not feel like a second checkpoint.

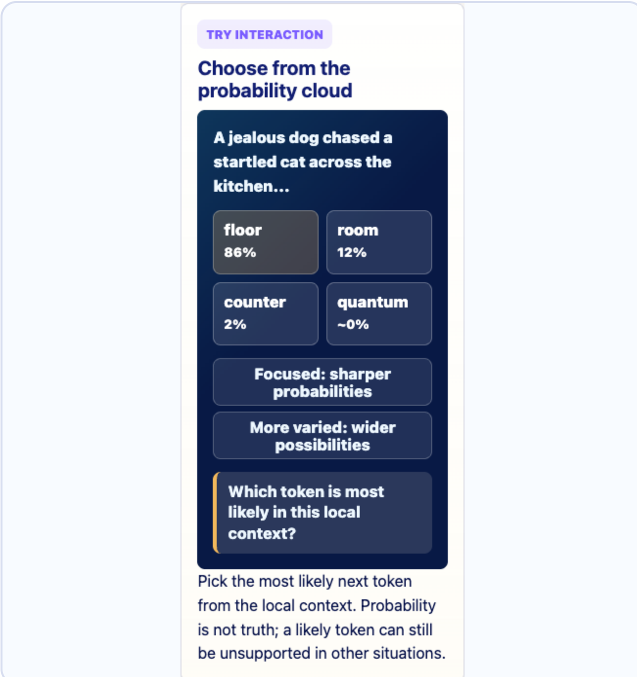
Visual duplicate: Watch during human testing; it may reinforce the visual closely.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

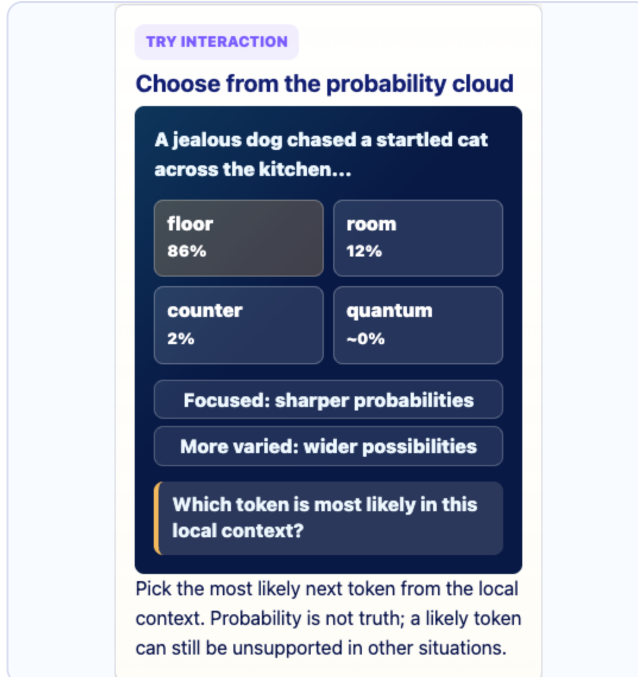
Rubric average: 4.1/5

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.



320px



390px

23. Autoregression

Stage: The Day Repeats

Learning objective: Autoregression is response generation by repeatedly choosing one next token, appending it to the context, and running the model again. Practice boundary: The model writes the whole response at once.

Exercise: Tap-step append loop ordered sequence before P3 after P3

Prompt: Tap through choose token, append, and run again to see how the response-so-far grows.

Expected action: Step through choose token, append, and run again.

Items / choices / nodes / steps: choose token | append | run again

Target behavior: Complete the append-and-repeat loop.

Feedback behavior: Append-and-repeat feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The sequence is the mechanism.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

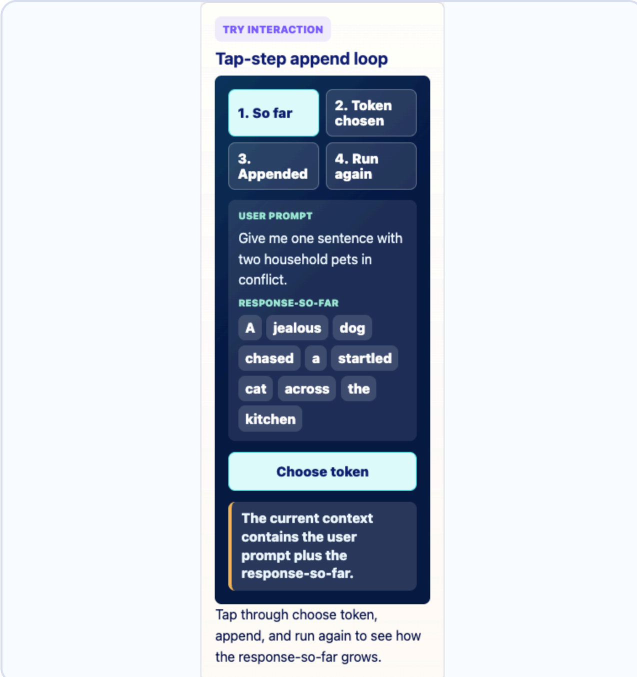
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

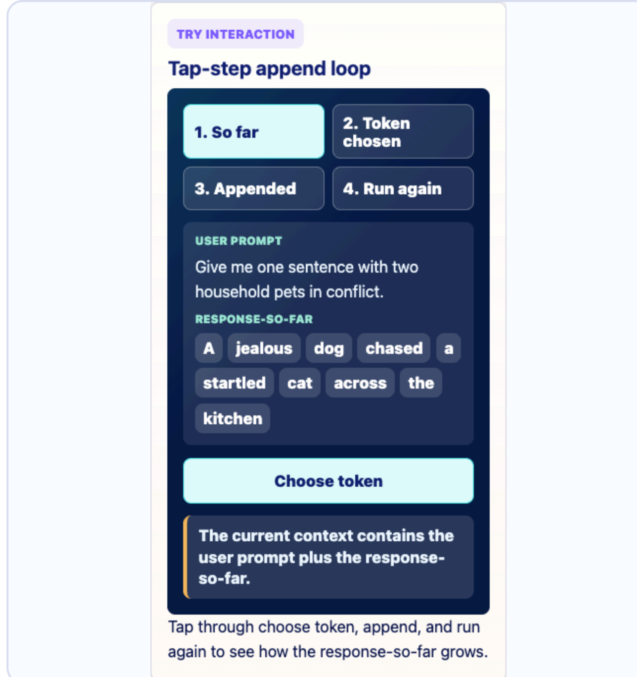
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.7/5

No P0/P1 issue remains.


TRY INTERACTION
Tap-step append loop
1. So far
2. Token chosen
3. Appended
4. Run again
USER PROMPT
Give me one sentence with two household pets in conflict.
RESPONSE-SO-FAR
A
jealous
dog
chased
a
startled
cat
across
the
kitchen
Choose token
The current context contains the user prompt plus the response-so-far.
Tap through choose token, append, and run again to see how the response-so-far grows.

320px


TRY INTERACTION
Tap-step append loop
1. So far
2. Token chosen
3. Appended
4. Run again
USER PROMPT
Give me one sentence with two household pets in conflict.
RESPONSE-SO-FAR
A
jealous
dog
chased
a
startled
cat
across
the
kitchen
Choose token
The current context contains the user prompt plus the response-so-far.
Tap through choose token, append, and run again to see how the response-so-far grows.

390px

24. Context Window

Stage: The Day Repeats

Learning objective: The context window is the limited amount of tokens or media the model can consider at one time. Practice boundary: The context window is permanent memory.

Exercise: Push the context items card stack / tray before P1 after P3

Prompt: Tap items into a four-slot window and watch the oldest item fall out.

Expected action: Push items into a bounded context tray.

Items / choices / nodes / steps: old message | system instruction | user prompt | retrieved note | response so far

Target behavior: Observe the oldest item fall out after the window fills.

Feedback behavior: Temporary-window feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The fixed order makes the capacity limit visible.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

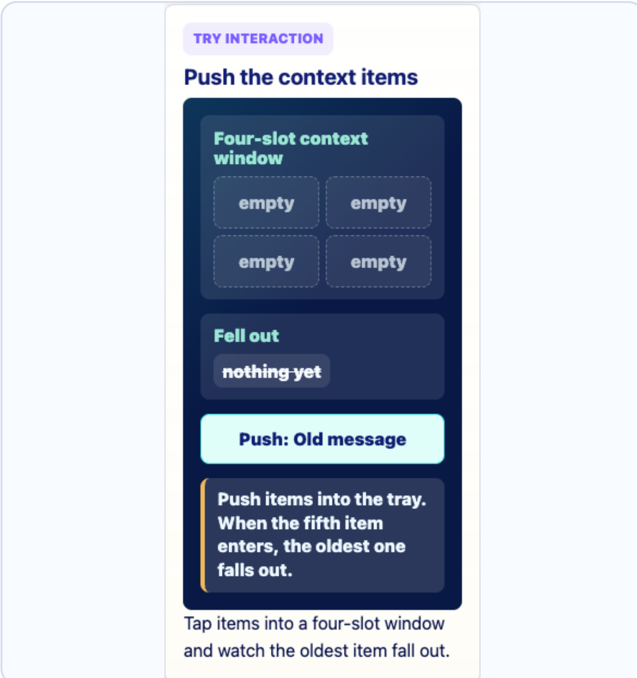
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

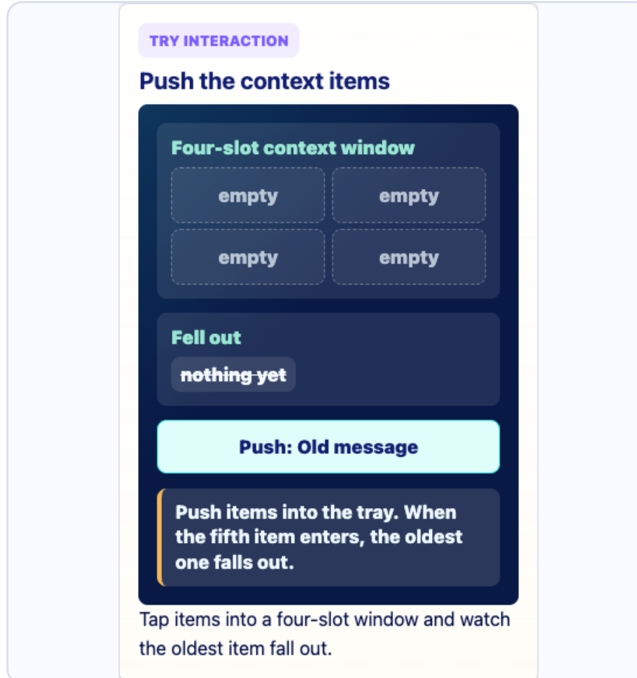
Rubric average: 4.7/5

Fixed: Changed ambiguous “card” wording to “item” in the Journey copy and live context-tray feedback.

No P0/P1 issue remains.


TRY INTERACTION
Push the context items
Four-slot context window
empty
empty
empty
empty
Fell out
~~nothing yet~~
Push: Old message
Push items into the tray. When the fifth item enters, the oldest one falls out.
Tap items into a four-slot window and watch the oldest item fall out.

320px


TRY INTERACTION
Push the context items
Four-slot context window
empty
empty
empty
empty
Fell out
~~nothing yet~~
Push: Old message
Push items into the tray. When the fifth item enters, the oldest one falls out.
Tap items into a four-slot window and watch the oldest item fall out.

390px

25. RAG and Retrieval

Stage: The Day Repeats

Learning objective: Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's current context before the model generates a response. Practice boundary: RAG is training, permanent memory, guaranteed truth, or consciousness.

Exercise: Trace the retrieval lanes **tap-to-cycle** **before P3** **after P3**

Prompt: Tap prompt, retriever, notes, context tray, and response to see retrieval plus context, not training.

Expected action: Trace prompt, retriever, retrieved notes, context tray, and response.

Items / choices / nodes / steps: prompt | retriever | retrieved notes | context tray | response

Target behavior: See retrieval plus context, not training.

Feedback behavior: Retrieval-plus-context feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: RAG has a stepwise system path.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

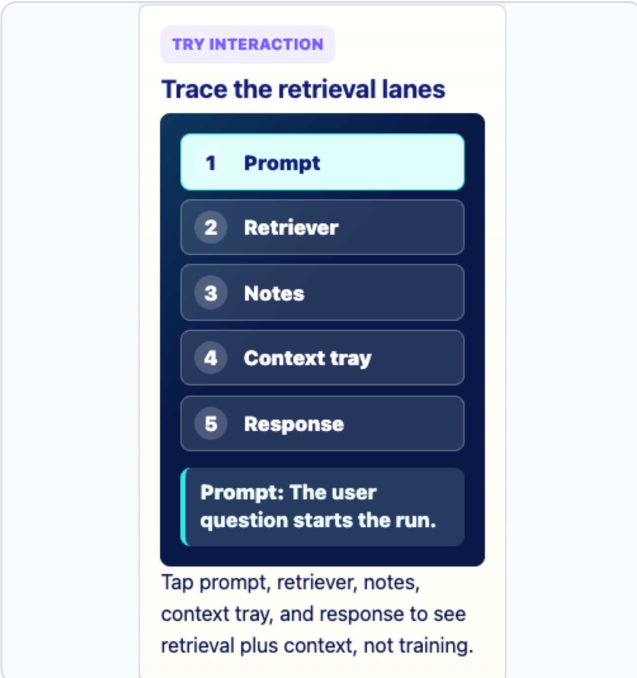
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

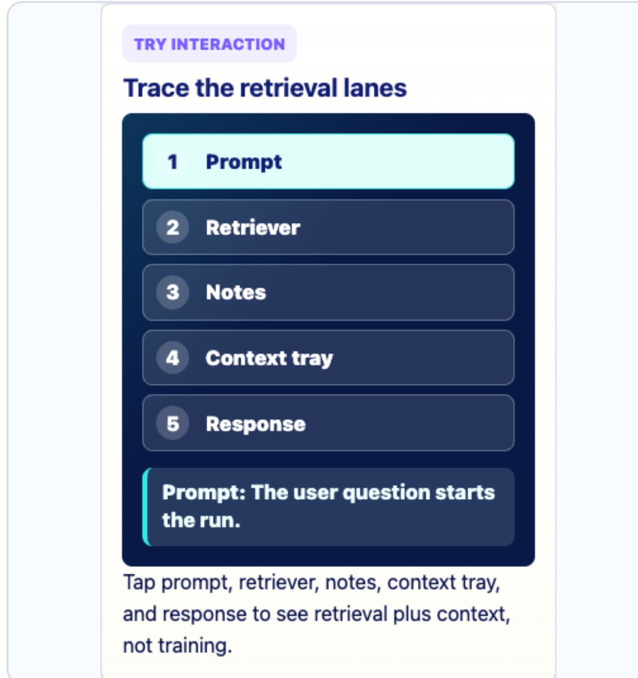
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Trace the retrieval lanes
1 Prompt
2 Retriever
3 Notes
4 Context tray
5 Response
Prompt: The user question starts the run.
Tap prompt, retriever, notes, context tray, and response to see retrieval plus context, not training.

320px


TRY INTERACTION
Trace the retrieval lanes
1 Prompt
2 Retriever
3 Notes
4 Context tray
5 Response
Prompt: The user question starts the run.
Tap prompt, retriever, notes, context tray, and response to see retrieval plus context, not training.

390px

26. Grounding

Stage: The Day Repeats

Learning objective: Grounding means connecting a model’s response to available evidence, such as retrieved documents, citations, data, or tool results. Practice boundary: A citation-looking answer is automatically grounded.

Exercise: Match claims to evidence connect node before P3 after P3

Prompt: Connect supported claims to evidence and leave unsupported claims unanchored.

Expected action: Connect claims to supporting evidence.

Items / choices / nodes / steps: supported claim | evidence | unsupported claim

Target behavior: Connect supported claims and leave unsupported claims unanchored.

Feedback behavior: Evidence-match feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed examples prevent source confusion.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

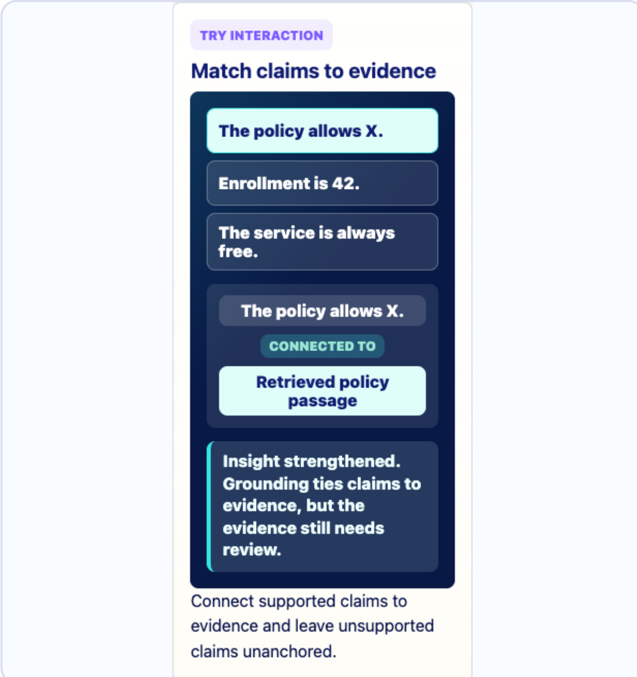
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

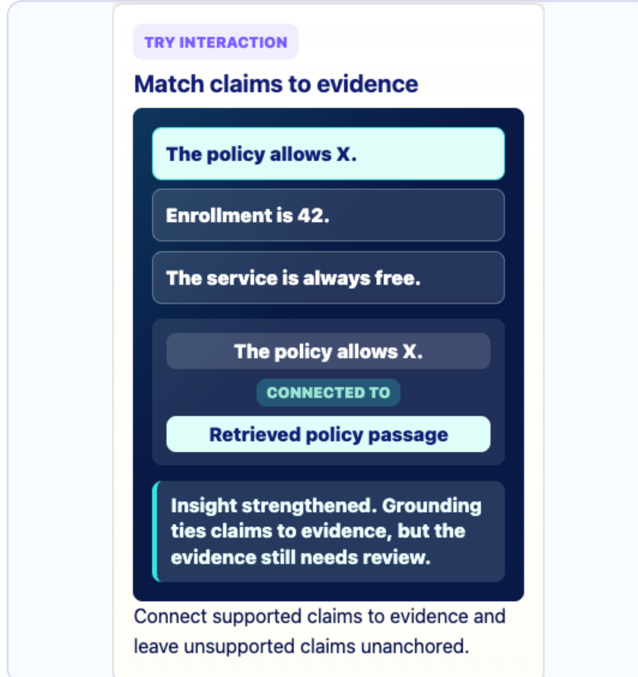
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Match claims to evidence
The policy allows X.
Enrollment is 42.
The service is always free.
The policy allows X.
CONNECTED TO
Retrieved policy passage
Insight strengthened. Grounding ties claims to evidence, but the evidence still needs review.
Connect supported claims to evidence and leave unsupported claims unanchored.

320px


TRY INTERACTION
Match claims to evidence
The policy allows X.
Enrollment is 42.
The service is always free.
The policy allows X.
CONNECTED TO
Retrieved policy passage
Insight strengthened. Grounding ties claims to evidence, but the evidence still needs review.
Connect supported claims to evidence and leave unsupported claims unanchored.

390px

27. Hallucinations

Stage: The Day Repeats

Learning objective: A hallucination is a fluent model output that is unsupported, false, fabricated, or not grounded in the available evidence. Practice boundary: A hallucination means the model is lying or malicious.

Exercise: Mark missing supports single choice before P3 after P3

Prompt: Read three fluent claims and mark the ones that need evidence.

Expected action: Choose the fluent claim that needs evidence before trust.

Items / choices / nodes / steps: supported policy claim | unsupported date claim | review-needed caveat

Target behavior: Select the unsupported fluent claim.

Feedback behavior: Unsupported-output feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The contrast depends on stable evidence labels.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: Partial risk: watch during human testing so it does not feel like a second checkpoint.

Visual duplicate: No serious duplicate; it adds a learner action.

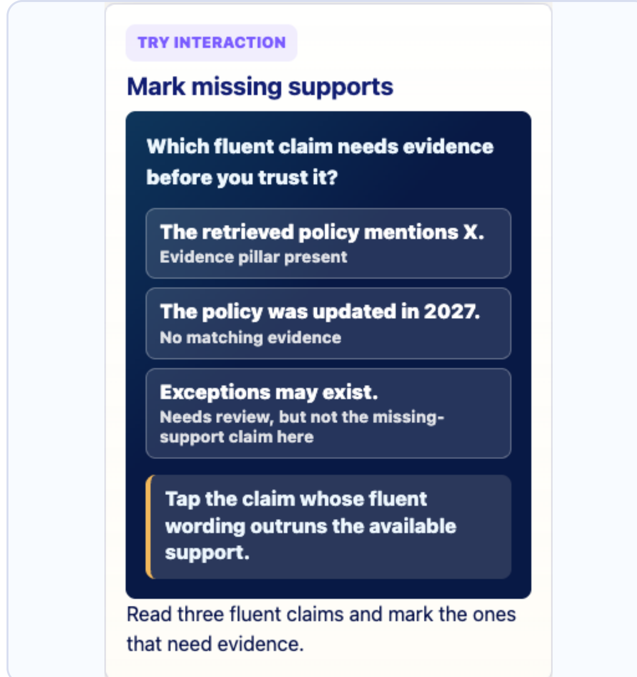
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Mark missing supports
Which fluent claim needs evidence before you trust it?
The retrieved policy mentions X.
Evidence pillar present
The policy was updated in 2027.
No matching evidence
Exceptions may exist.
Needs review, but not the missing-support claim here
Tap the claim whose fluent wording outruns the available support.
Read three fluent claims and mark the ones that need evidence.

320px


TRY INTERACTION
Mark missing supports
Which fluent claim needs evidence before you trust it?
The retrieved policy mentions X.
Evidence pillar present
The policy was updated in 2027.
No matching evidence
Exceptions may exist.
Needs review, but not the missing-support claim here
Tap the claim whose fluent wording outruns the available support.
Read three fluent claims and mark the ones that need evidence.

390px

28. How AI Learns

Stage: Twilight: The Wider Landscape

Learning objective: AI systems can change or behave differently through durable training, human feedback, retrieval, prompting, or temporary in-context steering. Practice boundary: Any useful response means the model permanently learned something.

Exercise: Sort the learning modes sort / group before P1 after P3

Prompt: Durable update, temporary steering, retrieval, and evaluation solve different problems.

Expected action: Sort durable update, temporary steering, retrieval, and evaluation modes.

Items / choices / nodes / steps: pretraining | prompting | RAG | evaluation

Target behavior: Assign each learning or steering mode to the right mechanism.

Feedback behavior: What-changed feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable options preserve the mechanism map.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

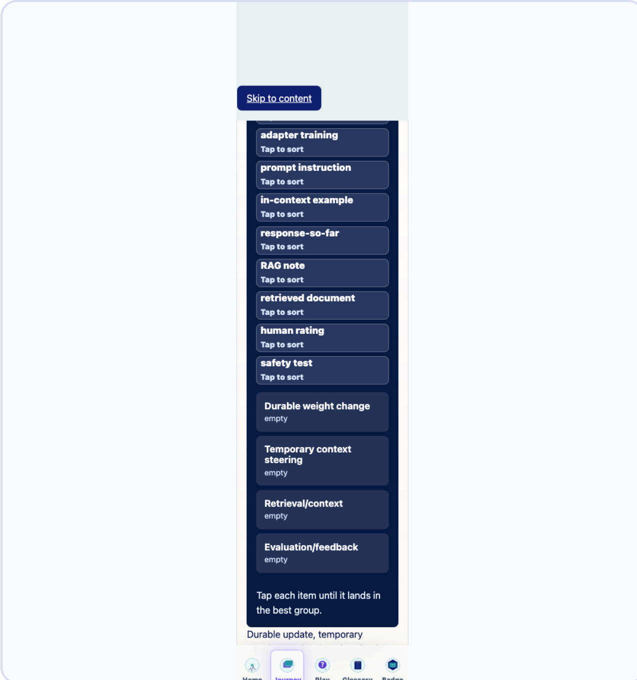
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

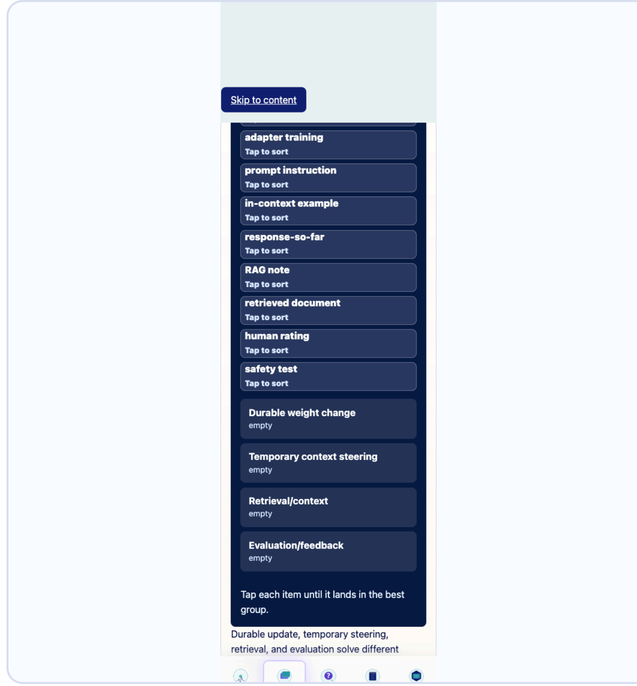
Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.


Skip to content
adapter training
Tap to sort
prompt instruction
Tap to sort
in-context example
Tap to sort
response-so-far
Tap to sort
RAG note
Tap to sort
retrieved document
Tap to sort
human rating
Tap to sort
safety test
Tap to sort
Durable weight change
empty
Temporary context steering
empty
Retrieval/context
empty
Evaluation/feedback
empty
Tap each item until it lands in the best group.
Durable update, temporary
Home
Journey
Play
Glossary
Badge

320px


Skip to content
adapter training
Tap to sort
prompt instruction
Tap to sort
in-context example
Tap to sort
response-so-far
Tap to sort
RAG note
Tap to sort
retrieved document
Tap to sort
human rating
Tap to sort
safety test
Tap to sort
Durable weight change
empty
Temporary context steering
empty
Retrieval/context
empty
Evaluation/feedback
empty
Tap each item until it lands in the best group.
Durable update, temporary steering,
retrieval, and evaluation solve different

390px

29. Diffusion vs Autoregression

Stage: Twilight: The Wider Landscape

Learning objective: Autoregressive models generate one token at a time, while diffusion models often generate by gradually denoising a noisy pattern. Practice boundary: All generative AI works like ChatGPT.

Exercise: Step through two loops **comparison toggle** **before P3** **after P3**

Prompt: Tap Token path or Denoise path to compare append-and-repeat with denoise-and-refine.

Expected action: Step through token append path and denoise path.

Items / choices / nodes / steps: token path | denoise path

Target behavior: Compare autoregressive text with diffusion denoising.

Feedback behavior: Autoregression-versus-diffusion feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The comparison follows two named paths.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

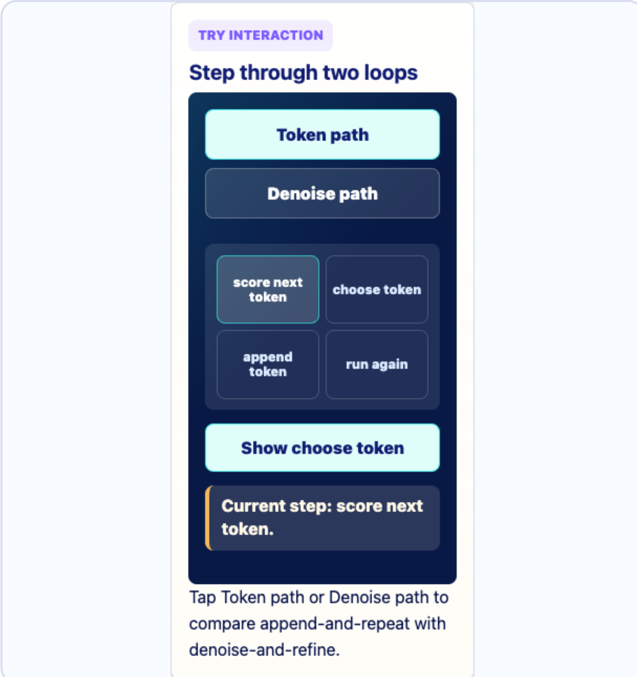
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

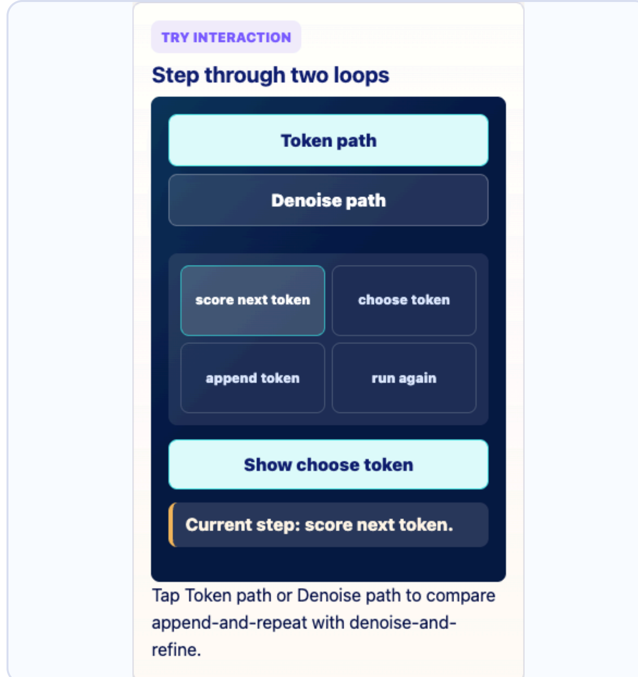
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.9/5

No P0/P1 issue remains.


TRY INTERACTION
Step through two loops
Token path
Denoise path
score next token
choose token
append token
run again
Show choose token
Current step: score next token.
Tap Token path or Denoise path to compare append-and-repeat with denoise-and-refine.

320px


TRY INTERACTION
Step through two loops
Token path
Denoise path
score next token
choose token
append token
run again
Show choose token
Current step: score next token.
Tap Token path or Denoise path to compare append-and-repeat with denoise-and-refine.

390px

30. Multimodal AI

Stage: Twilight: The Wider Landscape

Learning objective: Multimodal AI works with more than one kind of input or output, such as text, images, audio, video, or code. Practice boundary: Multimodal AI has human-like perception or feelings.

Exercise: Match media lanes mixed / other before P3 after P3

Prompt: Tap an input-output pair to see how different media can be represented or connected.

Expected action: Tap an input-output media pair.

Items / choices / nodes / steps: image to caption | audio to transcript | text to image

Target behavior: See different media represented or connected together.

Feedback behavior: Representation-not-human-senses feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed pairs keep multimodal examples concrete.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

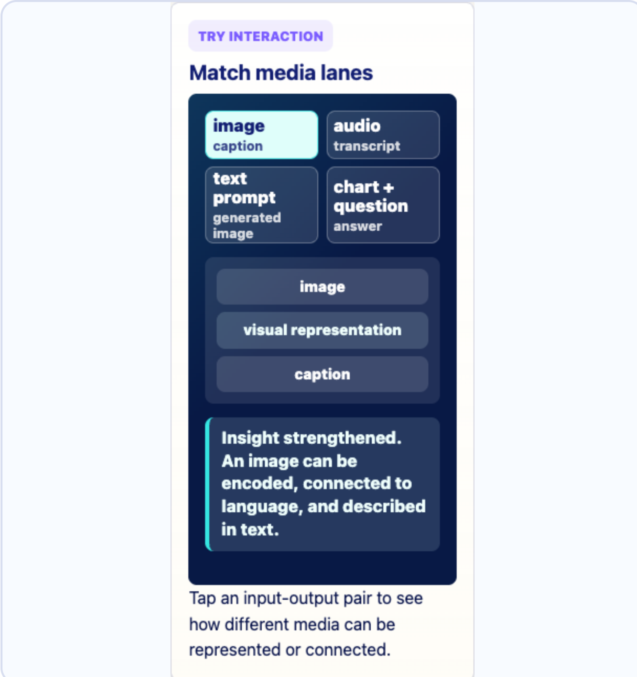
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.7/5

No P0/P1 issue remains.


TRY INTERACTION
Match media lanes
image
caption
audio
transcript
text prompt
generated image
chart + question
answer
image
visual representation
caption
Insight strengthened. An image can be encoded, connected to language, and described in text.
Tap an input-output pair to see how different media can be represented or connected.

320px


TRY INTERACTION
Match media lanes
image
caption
audio
transcript
text prompt
generated image
chart + question
answer
image
visual representation
caption
Insight strengthened. An image can be encoded, connected to language, and described in text.
Tap an input-output pair to see how different media can be represented or connected.

390px

31. The Perfect Storm

Stage: Twilight: The Wider Landscape

Learning objective: Modern LLMs became possible when human-created digital data, powerful compute and storage, deep-learning methods, human labor, and economic incentives converged. Practice boundary: LLMs appeared because of one event.

Exercise: Tap storm ingredients tap-to-cycle before P3 after P3

Prompt: Tap each stream to see why capability came from many converging conditions, not one mysterious spark.

Expected action: Tap converging ingredient streams.

Items / choices / nodes / steps: data | compute | storage | methods | labor | incentives

Target behavior: Reveal convergence rather than one spark.

Feedback behavior: Convergence-not-spark feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The ingredient list is the concept map.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

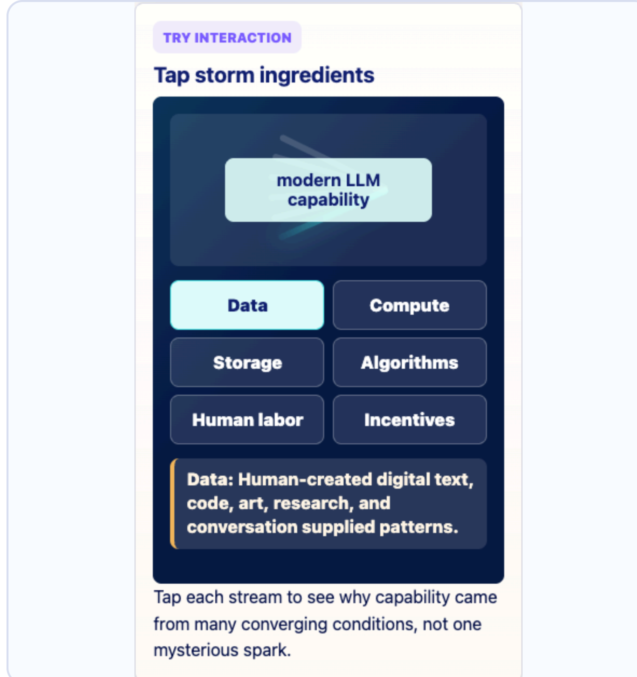
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Tap storm ingredients
modern LLM capability
Data
Compute
Storage
Algorithms
Human labor
Incentives
Data: Human-created digital text, code, art, research, and conversation supplied patterns.
Tap each stream to see why capability came from many converging conditions, not one mysterious spark.

320px


TRY INTERACTION
Tap storm ingredients
modern LLM capability
Data
Compute
Storage
Algorithms
Human labor
Incentives
Data: Human-created digital text, code, art, research, and conversation supplied patterns.
Tap each stream to see why capability came from many converging conditions, not one mysterious spark.

390px

32. Collective Intelligence, Extracted

Stage: Midnight Ledger

Learning objective: Generative AI depends on patterns learned from human-created language, art, code, documents, research, and culture. Practice boundary: The model created its abilities alone.

Exercise: Which questions stay human? sort / group before P1 after P2

Prompt: Sort rights and source questions away from model mechanics.

Expected action: Sort human-rights questions away from model mechanics.

Items / choices / nodes / steps: creator consent | source credit | token probability | weight update

Target behavior: Separate institutional questions from model mechanics.

Feedback behavior: Accountability feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed set keeps ethics and mechanics separate.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: Watch during human testing; it may reinforce the visual closely.

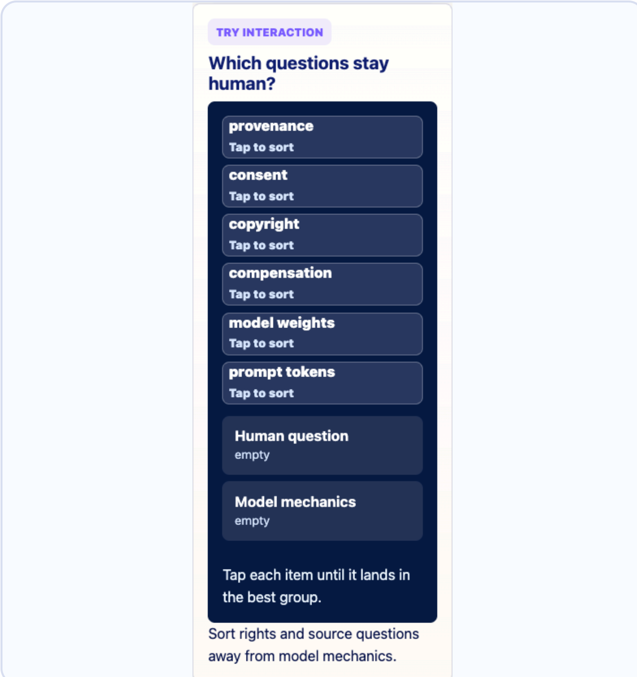
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.2/5

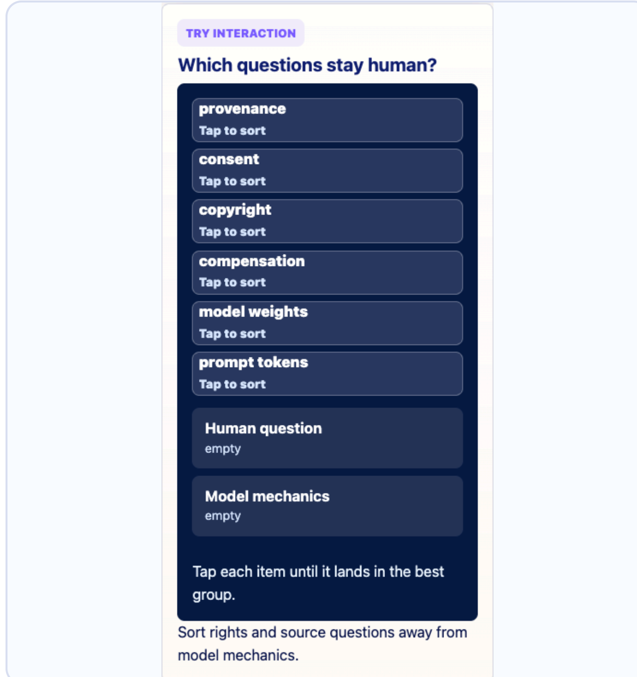
Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.



320px



390px

33. Costs We Must Count

Stage: Midnight Ledger

Learning objective: Generative AI has physical, social, cultural, and ethical costs that must be counted honestly. Practice boundary: Digital means weightless or cost-free.

Exercise: Tap the cost ledger tap-to-cycle before P3 after P3

Prompt: Tap a ledger entry to see what responsible AI literacy has to count.

Expected action: Tap cost ledger entries.

Items / choices / nodes / steps: energy | water | labor | infrastructure | maintenance

Target behavior: Name costs responsible AI literacy has to count.

Feedback behavior: Countable-cost feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Ledger order is not assessed.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

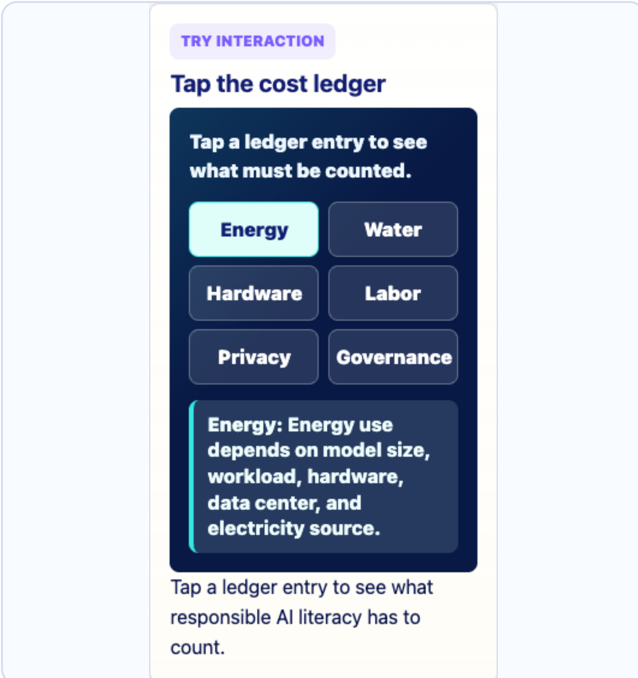
Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

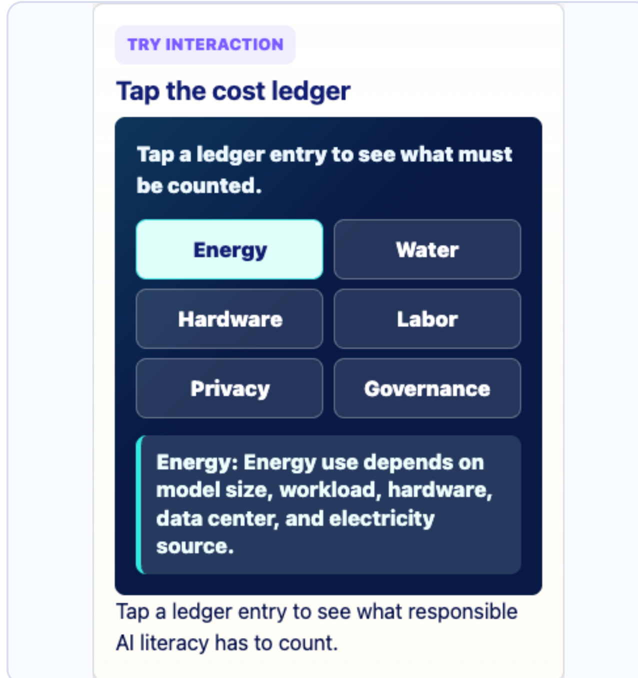
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Tap the cost ledger
Tap a ledger entry to see what must be counted.
Energy
Water
Hardware
Labor
Privacy
Governance
Energy: Energy use depends on model size, workload, hardware, data center, and electricity source.
Tap a ledger entry to see what responsible AI literacy has to count.

320px


TRY INTERACTION
Tap the cost ledger
Tap a ledger entry to see what must be counted.
Energy
Water
Hardware
Labor
Privacy
Governance
Energy: Energy use depends on model size, workload, hardware, data center, and electricity source.
Tap a ledger entry to see what responsible AI literacy has to count.

390px

34. Risk vs Myth

Stage: Midnight Ledger

Learning objective: Risk literacy means separating practical, mechanism-based AI risks from myths or unsupported claims about what models are. Practice boundary: Risk only means sci-fi doom, or every fluent model has a mind.

Exercise: Sort risk from myth sort / group before P1 after P3

Prompt: Sort each claim as a real mechanism risk or an unsupported story.

Expected action: Sort practical risks from unsupported stories.

Items / choices / nodes / steps: privacy leak | bad automation | model wants power | instant omniscience

Target behavior: Separate mechanism-backed risks from myths.

Feedback behavior: Mechanism-risk feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable claims support misconception repair.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

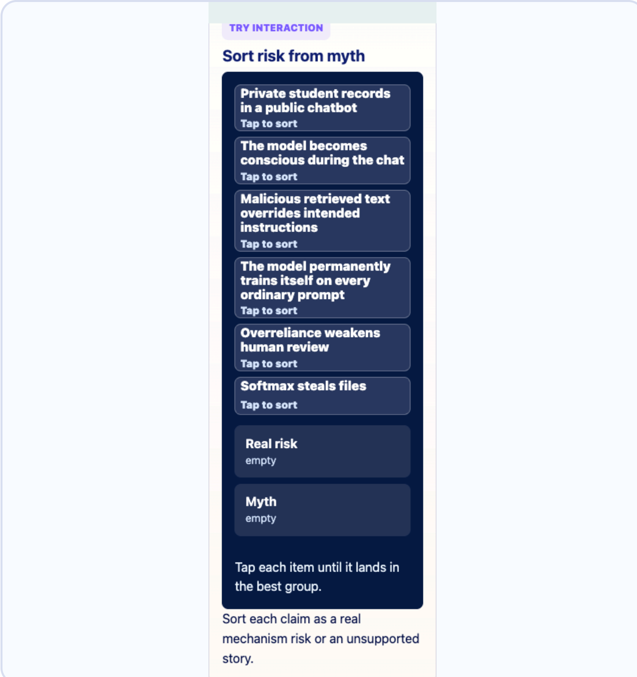
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

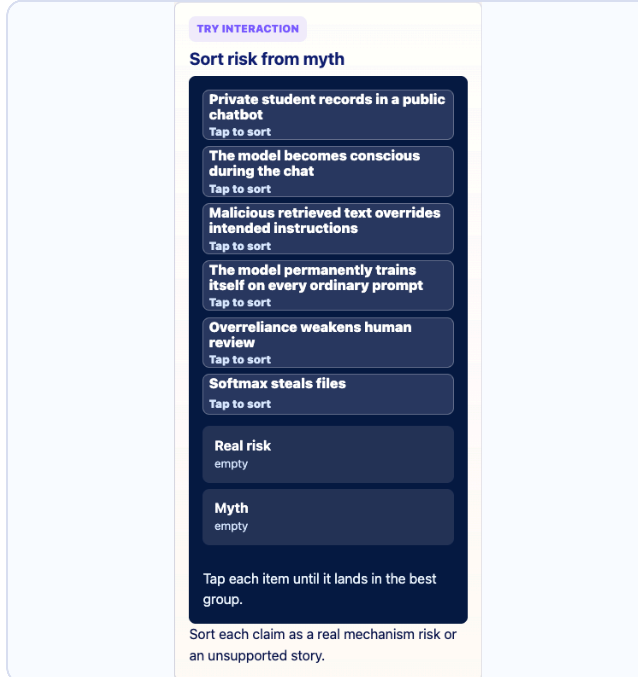
Rubric average: 4.7/5

Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.


TRY INTERACTION
Sort risk from myth
Private student records in a public chatbot
Tap to sort
The model becomes conscious during the chat
Tap to sort
Malicious retrieved text overrides intended instructions
Tap to sort
The model permanently trains itself on every ordinary prompt
Tap to sort
Overreliance weakens human review
Tap to sort
Softmax steals files
Tap to sort
Real risk
empty
Myth
empty
Tap each item until it lands in the best group.
Sort each claim as a real mechanism risk or an unsupported story.

320px


TRY INTERACTION
Sort risk from myth
Private student records in a public chatbot
Tap to sort
The model becomes conscious during the chat
Tap to sort
Malicious retrieved text overrides intended instructions
Tap to sort
The model permanently trains itself on every ordinary prompt
Tap to sort
Overreliance weakens human review
Tap to sort
Softmax steals files
Tap to sort
Real risk
empty
Myth
empty
Tap each item until it lands in the best group.
Sort each claim as a real mechanism risk or an unsupported story.

390px

35. Benefits Worth Taking Seriously

Stage: New Dawn

Learning objective: AI benefits are strongest when models amplify human judgment rather than replacing accountability. Practice boundary: AI benefits require believing in imminent utopia.

Exercise: Sort benefit strength sort / group before P1 after P2

Prompt: Sort benefit claims by confidence: useful now with review, plausible with safeguards, or speculative hype.

Expected action: Sort benefit claims by evidence strength.

Items / choices / nodes / steps: useful with review | plausible with safeguards | speculative hype

Target behavior: Classify benefits without hype.

Feedback behavior: Bounded-benefit feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The three tiers are the reasoning frame.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: Watch during human testing; it may reinforce the visual closely.

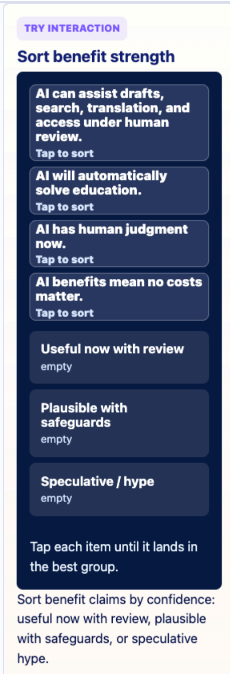
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.2/5

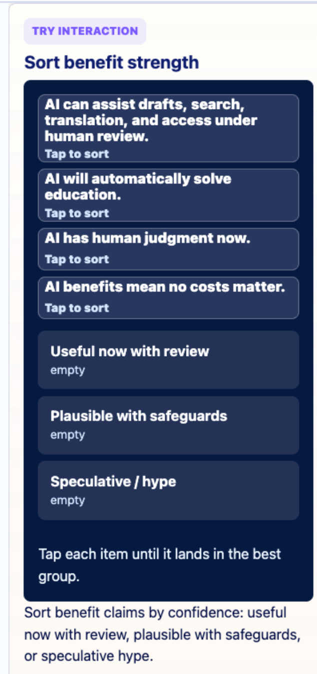
Fixed: Changed shared sort feedback from “cards” to “items” so sort/group interactions do not sound like Journey-card navigation.

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.



320px



390px

36. Human-Centered AI

Stage: New Dawn

Learning objective: Human-centered AI asks whether a system supports dignity, learning, responsibility, creativity, relationships, and the common good. Practice boundary: If AI is powerful, it should decide.

Exercise: Center the human purpose single choice before P3 after P3

Prompt: Choose the deployment that keeps people, review, and institutional accountability visible.

Expected action: Choose the more human-centered deployment.

Items / choices / nodes / steps: reviewable workflow | blind automation

Target behavior: Select the accountable deployment.

Feedback behavior: Review-and-accountability feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The two scenarios form a boundary contrast.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: Partial risk: watch during human testing so it does not feel like a second checkpoint.

Visual duplicate: No serious duplicate; it adds a learner action.

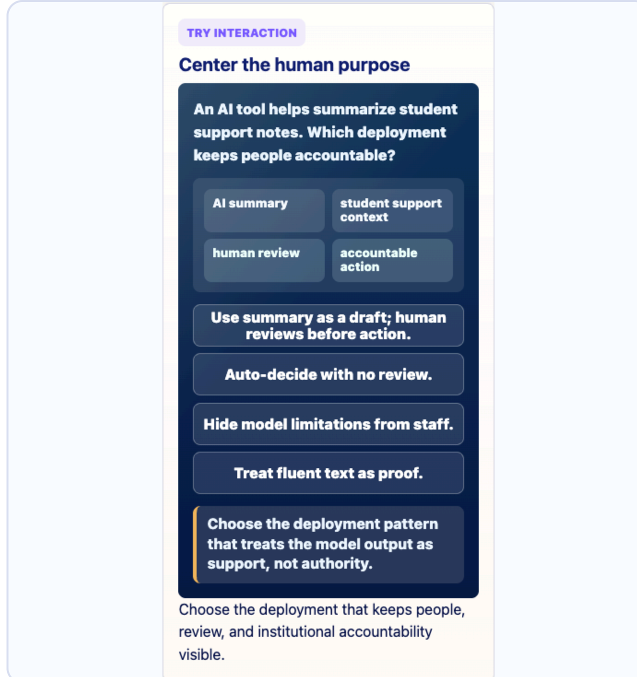
Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.8/5

No P0/P1 issue remains.


TRY INTERACTION
Center the human purpose
An AI tool helps summarize student support notes. Which deployment keeps people accountable?
AI summary
student support context
human review
accountable action
Use summary as a draft; human reviews before action.
Auto-decide with no review.
Hide model limitations from staff.
Treat fluent text as proof.
Choose the deployment pattern that treats the model output as support, not authority.
Choose the deployment that keeps people, review, and institutional accountability visible.

320px


TRY INTERACTION
Center the human purpose
An AI tool helps summarize student support notes. Which deployment keeps people accountable?
AI summary
student support context
human review
accountable action
Use summary as a draft; human reviews before action.
Auto-decide with no review.
Hide model limitations from staff.
Treat fluent text as proof.
Choose the deployment pattern that treats the model output as support, not authority.
Choose the deployment that keeps people, review, and institutional accountability visible.

390px

37. Better AI Is a Choice

Stage: New Dawn

Learning objective: The costs are shaped by design, deployment, governance, and incentives. Practice boundary: The harms are inevitable if the technology is useful.

Exercise: Choose the better path multi-select before P1 after P3

Prompt: Choose all design and governance levers that reduce risk or cost for an internal policy assistant.

Expected action: Choose all design and governance levers that reduce risk or cost.

Items / choices / nodes / steps: approved RAG | privacy | source review | task-fit model | monitoring | bad counterexamples

Target behavior: Select all useful levers and no risky counterexamples.

Feedback behavior: Selected lever feedback plus risk correction.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Fixed options keep multi-select review short.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4.7/5

Fixed: Added explicit "Choose all" language for the multi-select governance levers.

No P0/P1 issue remains.


Skip to content
better-AI levers.
Use RAG over approved documents
Protect sensitive data
Require source links and review
Choose a smaller task-fit model if adequate
Monitor errors
Upload private data to any chatbot
Assume bigger is always better
Remove human accountability
Ignore data provenance
Start with choices that protect data, use approved evidence, keep review, and fit the task.
Choose all design and governance levers that reduce risk or cost for an internal policy assistant.

320px


TRY INTERACTION
Choose the better path
A department wants an AI assistant for internal policy questions. Choose all better-AI levers.
Use RAG over approved documents
Protect sensitive data
Require source links and review
Choose a smaller task-fit model if adequate
Monitor errors
Upload private data to any chatbot
Assume bigger is always better
Remove human accountability
Ignore data provenance
Start with choices that protect data, use approved evidence, keep review, and fit the task.
Choose all design and governance levers that reduce risk or cost for an internal policy assistant.

390px

38. Effective Prompting from Model Literacy

Stage: New Dawn

Learning objective: Good prompts work because they shape the current context: task, constraints, examples, data, audience, evidence needs, uncertainty, and output format. Practice boundary: Prompting is special wording or permanent teaching.

Exercise: Pack the context tray **multi-select** **before P1** **after P3**

Prompt: Choose all missing prompt parts: task, audience/context, constraints, examples, evidence, uncertainty, format, and review cues.

Expected action: Choose all missing context parts for a stronger prompt.

Items / choices / nodes / steps: task | audience | constraints | examples | format | evidence | uncertainty | review

Target behavior: Add all missing prompt parts while keeping the task fixed.

Feedback behavior: Context-design feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: Stable parts reinforce prompt-as-context design.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

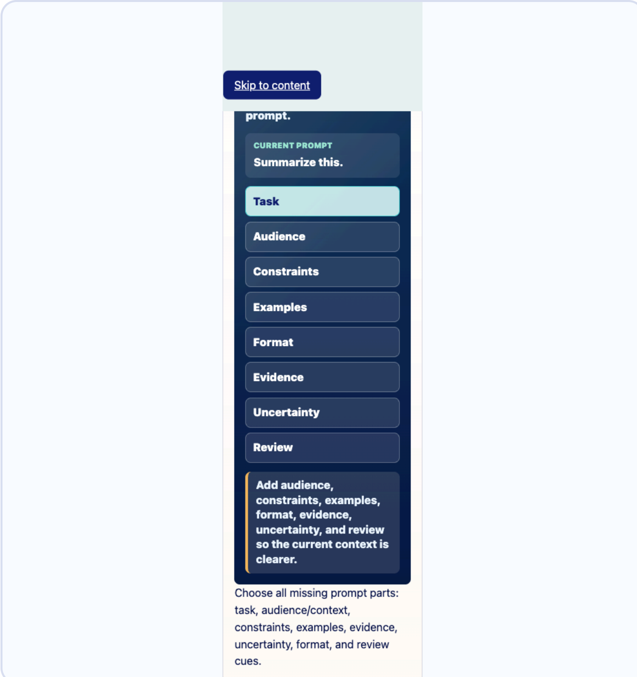
Visual duplicate: No serious duplicate; it adds a learner action.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

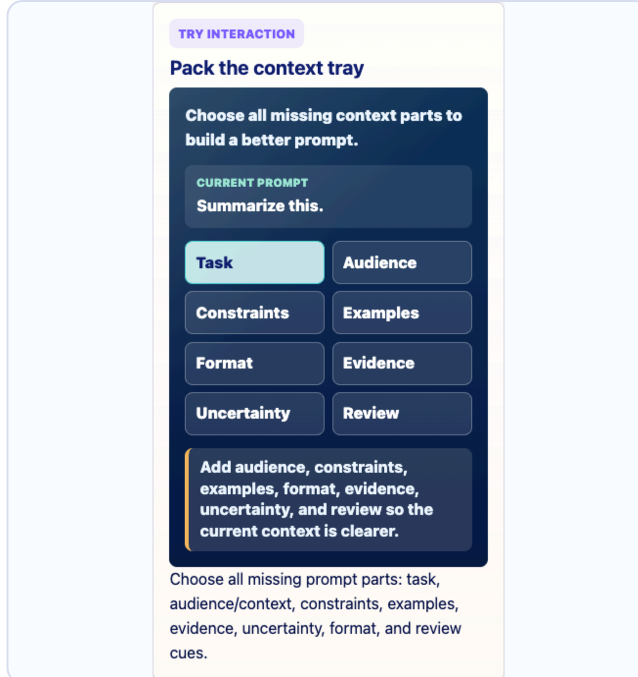
Rubric average: 4.7/5

Fixed: Added explicit “Choose all” language for the multi-select prompt-part picker.

No P0/P1 issue remains.


Skip to content
prompt.
CURRENT PROMPT
Summarize this.
Task
Audience
Constraints
Examples
Format
Evidence
Uncertainty
Review
Add audience, constraints, examples, format, evidence, uncertainty, and review so the current context is clearer.
Choose all missing prompt parts: task, audience/context, constraints, examples, evidence, uncertainty, format, and review cues.

320px


TRY INTERACTION
Pack the context tray
Choose all missing context parts to build a better prompt.
CURRENT PROMPT
Summarize this.
Task
Audience
Constraints
Examples
Format
Evidence
Uncertainty
Review
Add audience, constraints, examples, format, evidence, uncertainty, and review so the current context is clearer.
Choose all missing prompt parts: task, audience/context, constraints, examples, evidence, uncertainty, format, and review cues.

390px

39. Model Literate Synthesis

Stage: New Dawn

Learning objective: Model literacy means understanding both the mechanics and the human consequences. Practice boundary: Fluency equals understanding or wisdom.

Exercise: Light the map **ordered sequence** **before P2** **after P2**

Prompt: Put the model story in order, from prompt context to human accountability.

Expected action: Put the model-literacy chain in order.

Items / choices / nodes / steps: prompt | tokens | hidden states | logits | softmax | sample | append | human review

Target behavior: Order the mechanics-to-accountability chain.

Feedback behavior: Mechanics-to-accountability feedback.

Progress behavior: No. Journey progress is stored after checkpoint/reflection flow, not by the micro-interaction.

Randomization: No. Exercise controls use fixed order; checkpoint answers remain randomized elsewhere.

Fixed order: The synthesis is explicitly sequential.

Objective support: Yes. It practices the lesson mechanism or misconception boundary.

Checkpoint duplicate: No. It gives practice or reveal before checkpoint reasoning.

Visual duplicate: Watch during human testing; it may reinforce the visual closely.

Mobile notes: 320px no horizontal overflow; 390px no horizontal overflow; bottom nav does not cover controls after scroll

Rubric average: 4/5

No P0/P1 issue remains.

Human-test watch: Watch in human testing for abstraction level, redundancy, or replay value.


Skip to content
7 waiting
8 waiting
prompt enters context
text becomes tokens and IDs
embeddings become hidden states
logits become probabilities
one token is sampled
token is appended and the model runs again
RAG/grounding can add evidence
humans review and remain accountable
Reset chain
Begin with what enters the model run.
Put the model story in order, from prompt context to human accountability.

320px


Skip to content
4 waiting
5 waiting
6 waiting
7 waiting
8 waiting
prompt enters context
text becomes tokens and IDs
embeddings become hidden states
logits become probabilities
one token is sampled
token is appended and the model runs again
RAG/grounding can add evidence
humans review and remain accountable
Reset chain
Begin with what enters the model run.
Put the model story in order, from prompt context to human accountability.

390px